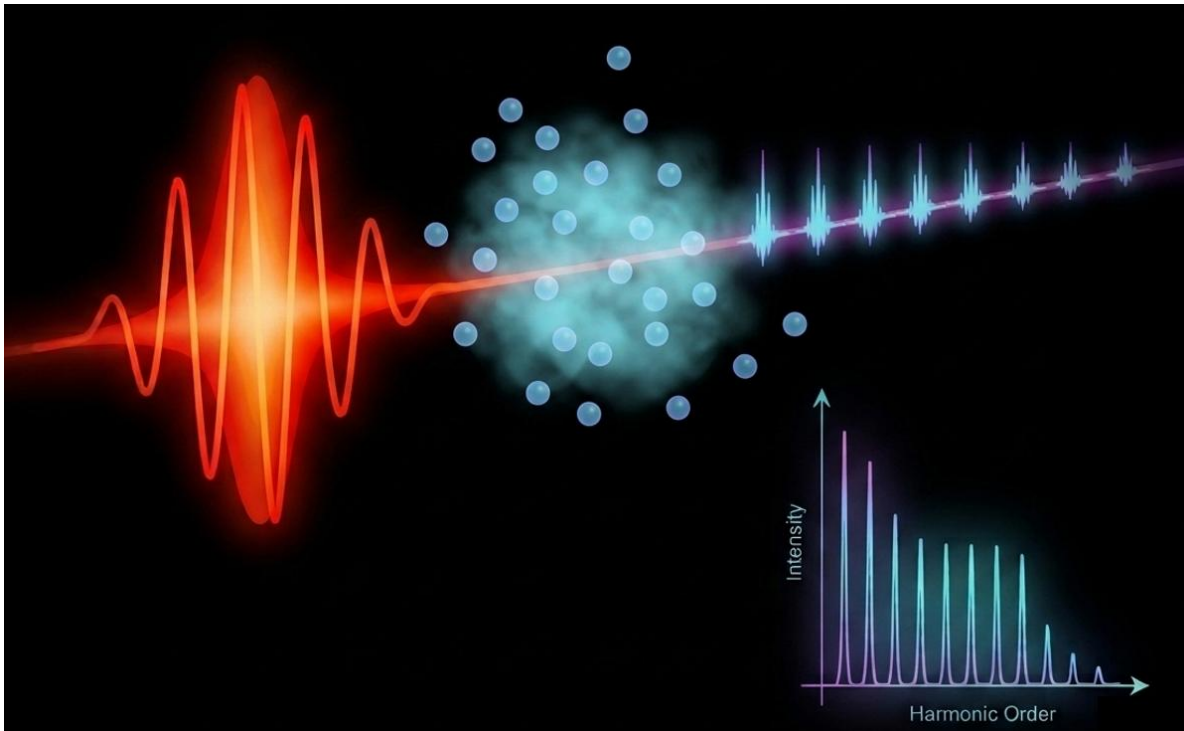


High harmonic generation and ultrafast electron dynamics simulations



IMPRS UFAST focus course

17th, 18th , 24th, 25th August
9:00-12:00h

Alba de las Heras
Theory department

UNTERSTÜTZT VON / SUPPORTED BY



Alexander von
HUMBOLDT
STIFTUNG



Max-Planck-Institut für
Struktur und Dynamik der Materie

Outline

Session 1: Fundamentals of Strong-Field Physics

Session 2: Microscopic modeling of laser-driven electron dynamics and spectroscopic signatures in high harmonic generation

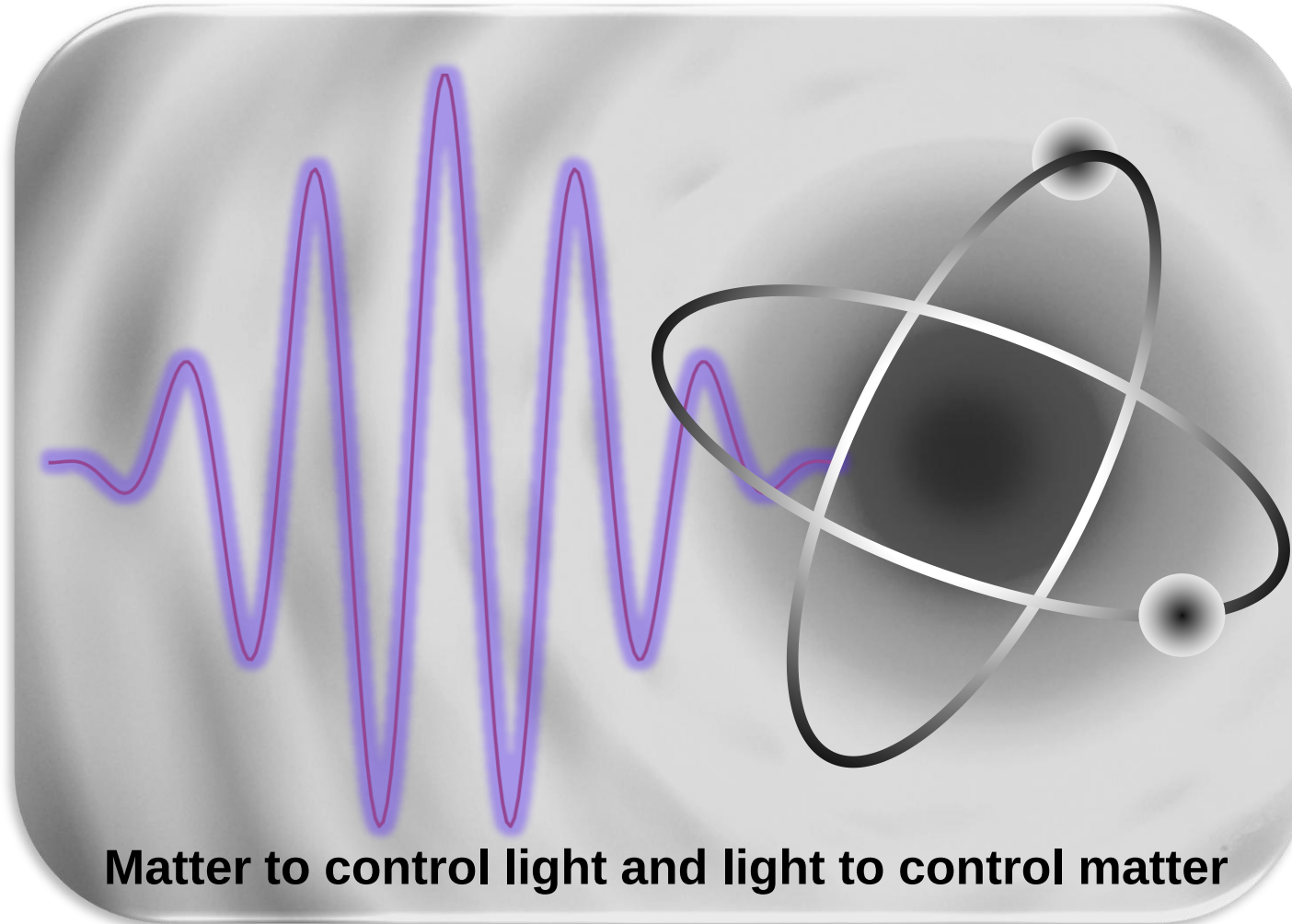
Session 3: Macroscopic picture of high harmonic generation and control of the properties of structured attosecond pulses

Session 4: Solid-state attosecond physics

Introduction to laser-matter interactions

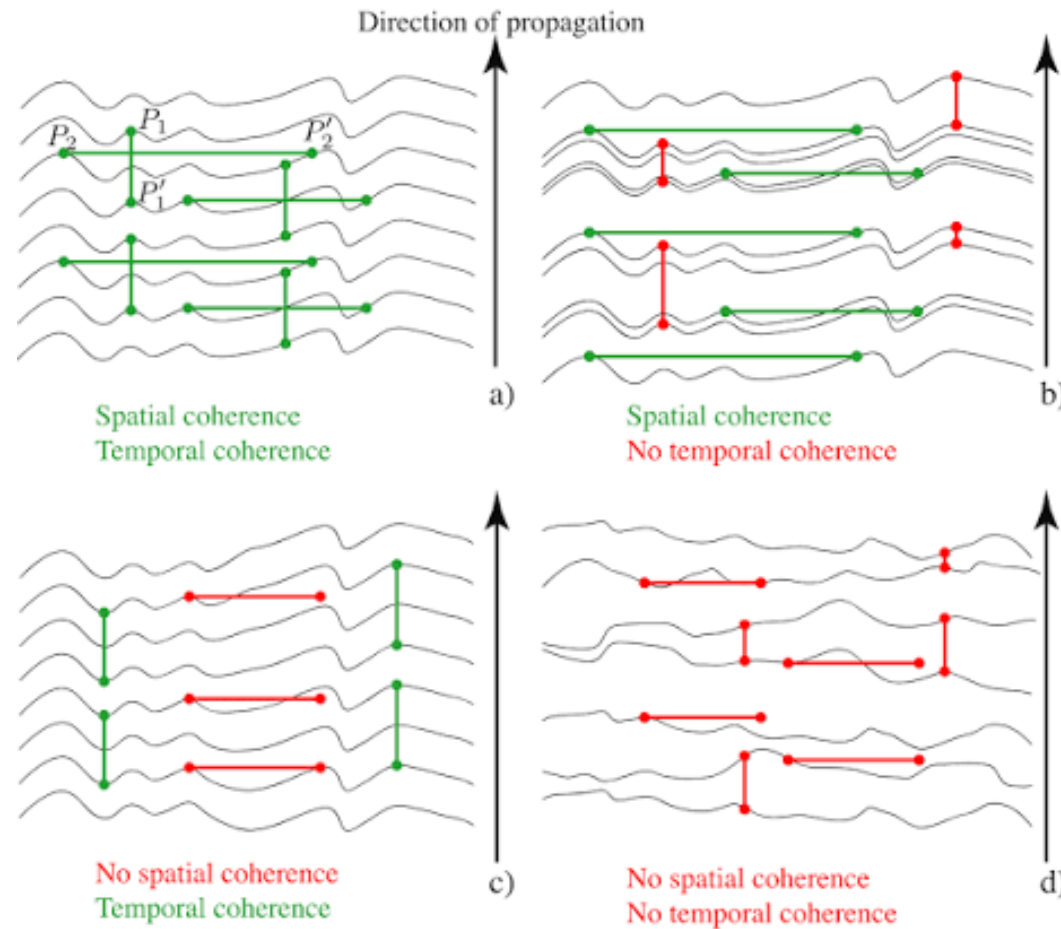
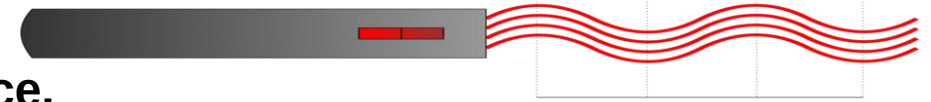
Why research on light-matter interactions?

A twofold opportunity for the control of the properties of light and the constituents of matter



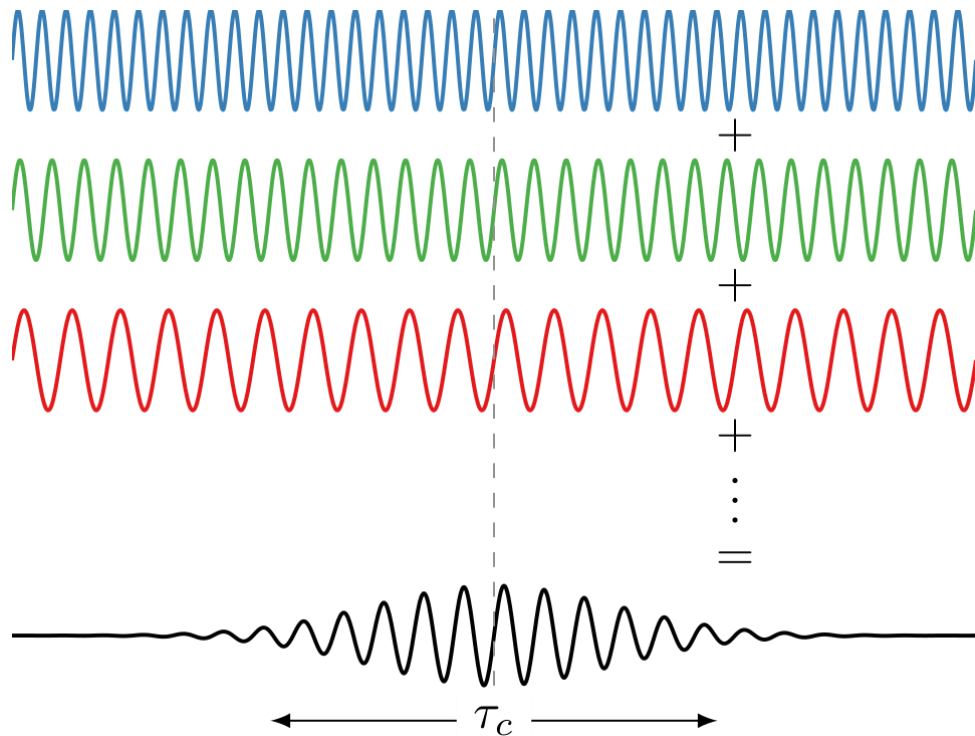
Among all the optical sources, why lasers?

- Lasers are coherent electromagnetic sources.
They exhibit temporal coherence and spatial coherence.

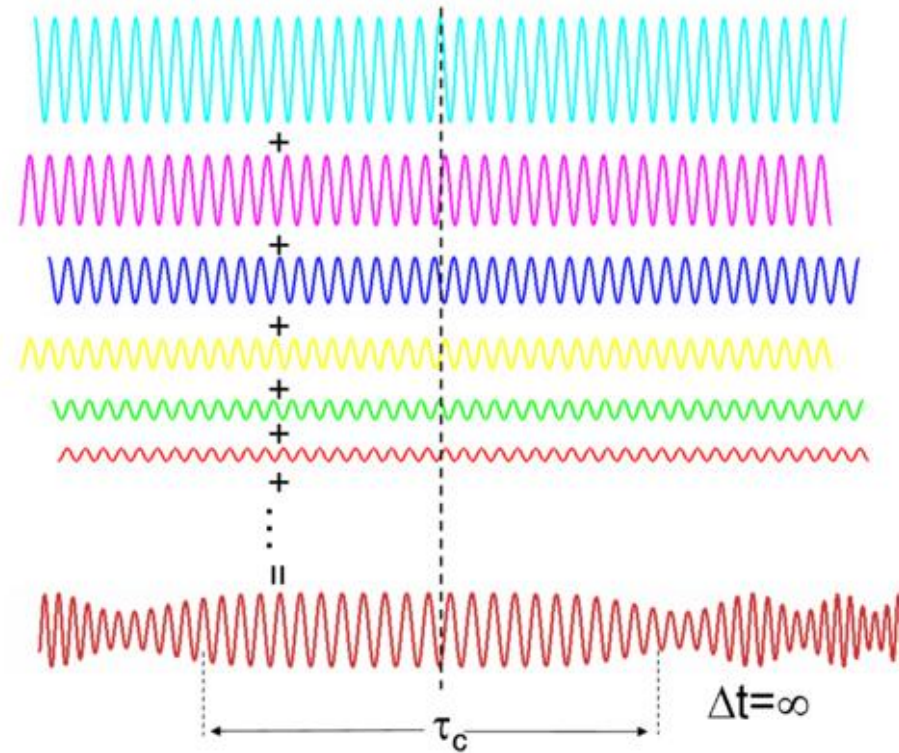


The concept of spectral coherence applies to multichromatic sources:

Spectral coherence



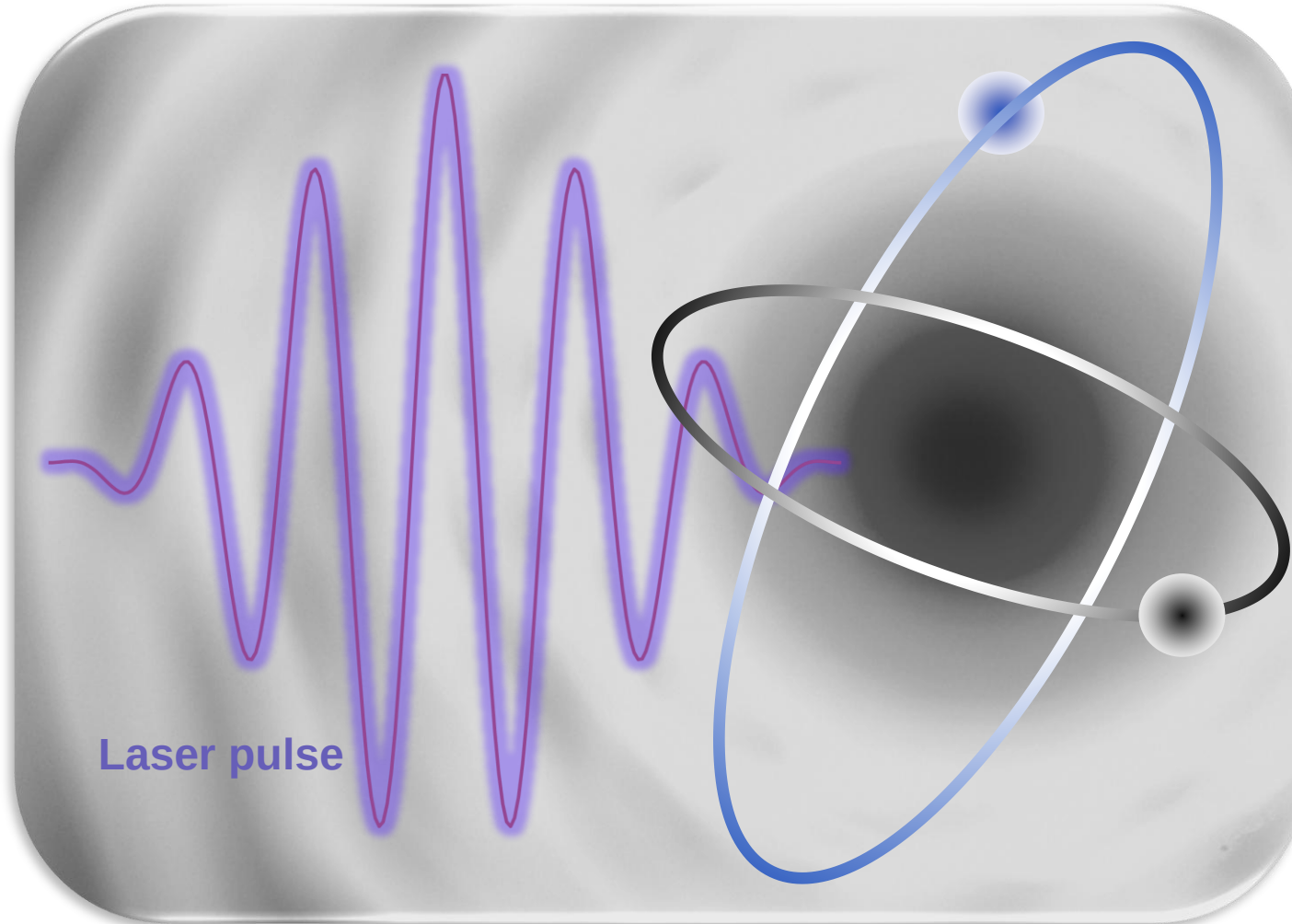
No spectral coherence



Fundamental laser-matter interaction process: **photoexcitation**

The quantum system is promoted to a higher energy state during the interaction with the laser

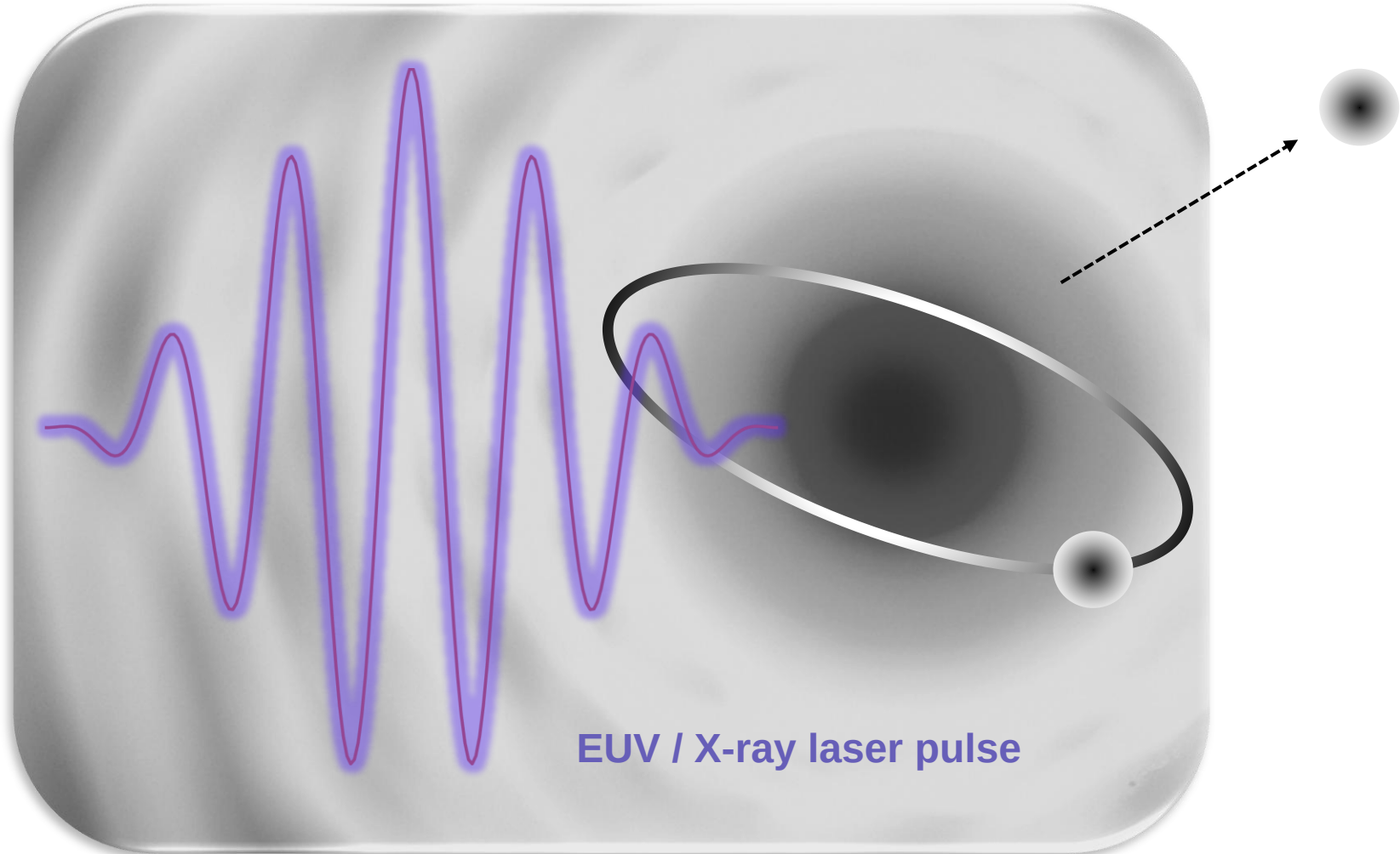
Single-photon
excitation
 $\hbar\omega_0 \geq E_f - E_i$



Fundamental laser-matter interaction process: photoionization

One or more electrons are released during the interaction with the laser

Single-photon
ionization
 $\hbar\omega_0 \geq I_p$



Linear Optics

- The optical response of matter is described as a linear response to the electromagnetic field.

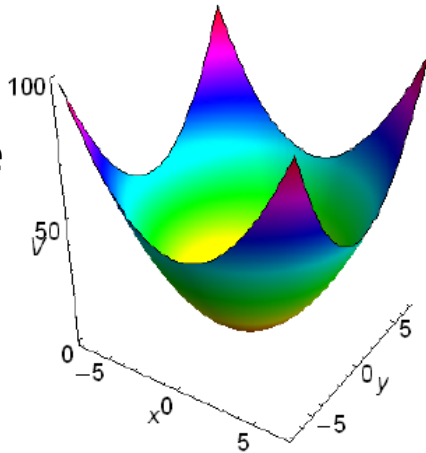
$$\mathbf{P}_\omega(\mathbf{E}_\omega) = \varepsilon_0 \chi(\omega) \mathbf{E}_\omega$$

$$\mathbf{J}_\omega(\mathbf{E}_\omega) = \sigma(\omega) \mathbf{E}_\omega$$

$$\mathbf{B}_\omega(\mathbf{H}_\omega) = \mu_0 \mathbf{H}_\omega$$

$$\mathbf{D}_\omega(\mathbf{E}_\omega) = \varepsilon_0 (1 + \chi(\omega)) \mathbf{E}_\omega = \varepsilon(\omega) \mathbf{E}_\omega$$

Classical Lorentz model describing the electron motion as a harmonic oscillator



Long times

$$\frac{d^2 x}{dt^2} = -\omega_0^2 x - \gamma \frac{dx}{dt} - \frac{e}{m} E_\omega e^{-i\omega t} \quad \Rightarrow \quad x(t) = \frac{e}{m} \frac{\omega^2 - \omega_0^2 - i\omega\gamma}{(\omega^2 - \omega_0^2)^2 + \omega^2\gamma^2} E_{\omega,x} e^{-i\omega t}$$

Macroscopic response:

$$P_\omega = -eN x_\omega \quad \Rightarrow \quad P = -eN x = - \underbrace{\frac{e^2 N}{m} \frac{\omega^2 - \omega_0^2 - i\omega\gamma}{(\omega^2 - \omega_0^2)^2 + \omega^2\gamma^2}}_{\varepsilon_0 \chi(\omega)} E e^{-i\omega t}$$



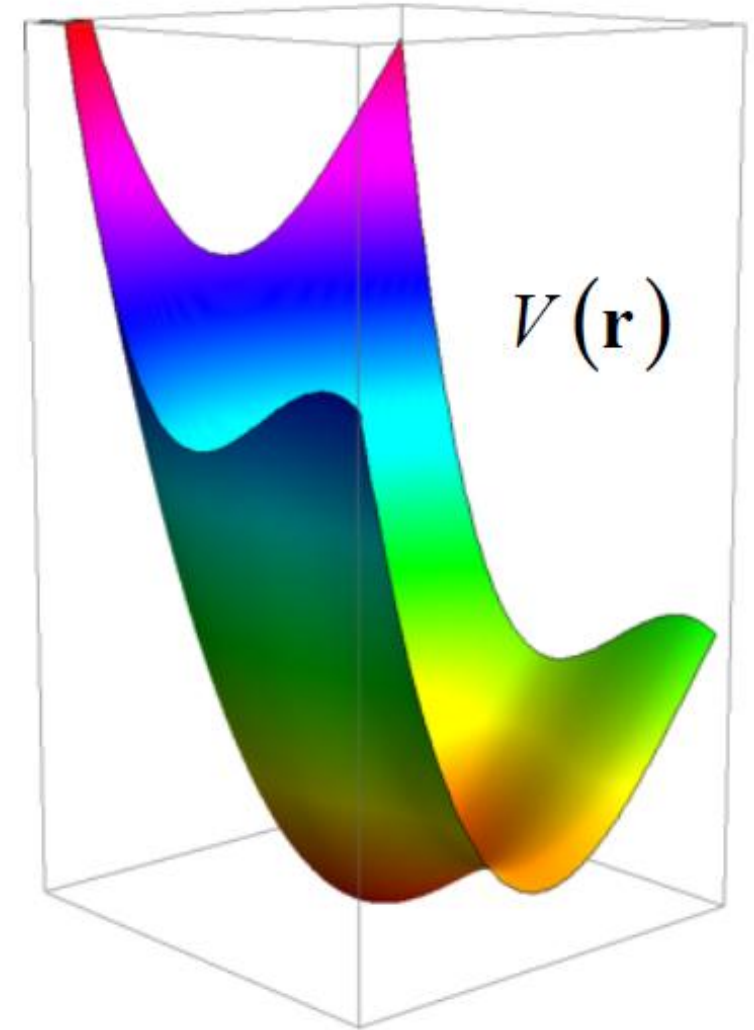
$$\mathbf{P}_\omega(\mathbf{E}_\omega) = \varepsilon_0 \chi(\omega) \mathbf{E}_\omega$$

Nonlinear optics

- The optical response of matter cannot be described as a linear response to the electromagnetic field, since higher-order terms become important.

$$\mathbf{P}(\mathbf{E}) = \varepsilon_0 \left[\chi^{(1)} \mathbf{E} + \chi^{(2)} \mathbf{E} \cdot \mathbf{E} + \chi^{(3)} \mathbf{E} \cdot \mathbf{E} \cdot \mathbf{E} + \dots \right]$$

- The standard linear superposition principle of electromagnetic waves does not apply, allowing different light waves to interact and alter one another inside the medium.
- The nonlinear optics response can arise in any medium under sufficiently high field amplitudes (typically around 10^{18} V/m)



Extension of the Lorentz model

$$m \frac{d^2 \mathbf{r}}{dt^2} = -\nabla V(\mathbf{r}) - m\gamma \frac{d\mathbf{r}}{dt} - e\mathbf{E}(t)$$

$$V(x) = V_0 + \left. \frac{dV}{dx} \right|_{x=0} x + \frac{1}{2} \left. \frac{d^2V}{dx^2} \right|_{x=0} x^2 + \frac{1}{6} \left. \frac{d^3V}{dx^3} \right|_{x=0} x^3 + \dots$$

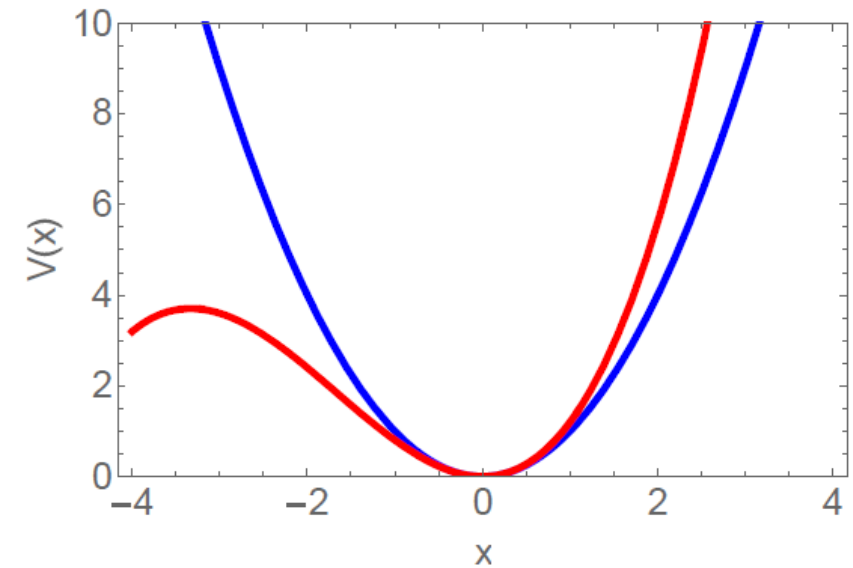
$$V(x) = \frac{1}{2} m \omega_0^2 x^2 + \frac{1}{3} m A_2 x^3 + \dots \quad \omega_0^2 = \left. \frac{1}{m} \frac{d^2V}{dx^2} \right|_{x=0} \quad A_2 = \left. \frac{1}{2m} \frac{d^3V}{dx^3} \right|_{x=0}$$

Perturbative solution:

$$x(t) = x_{\omega}^{(1)} e^{-i\omega t} + x_0^{(2)} + x_{\omega}^{(2)} e^{-i2\omega t} + \dots + cc$$

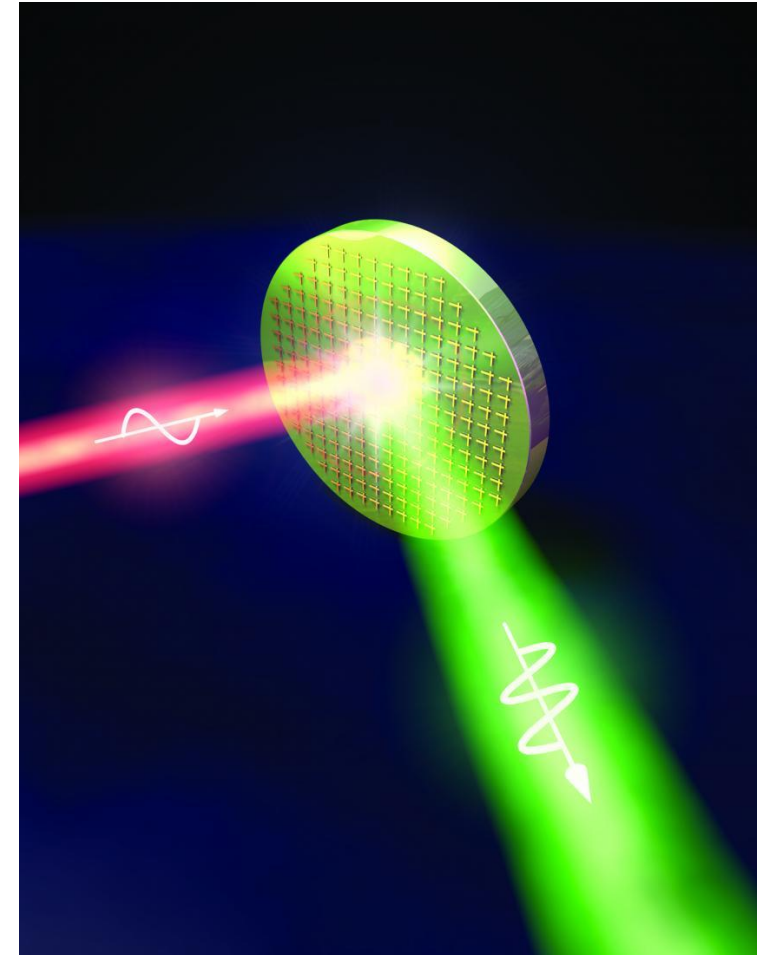
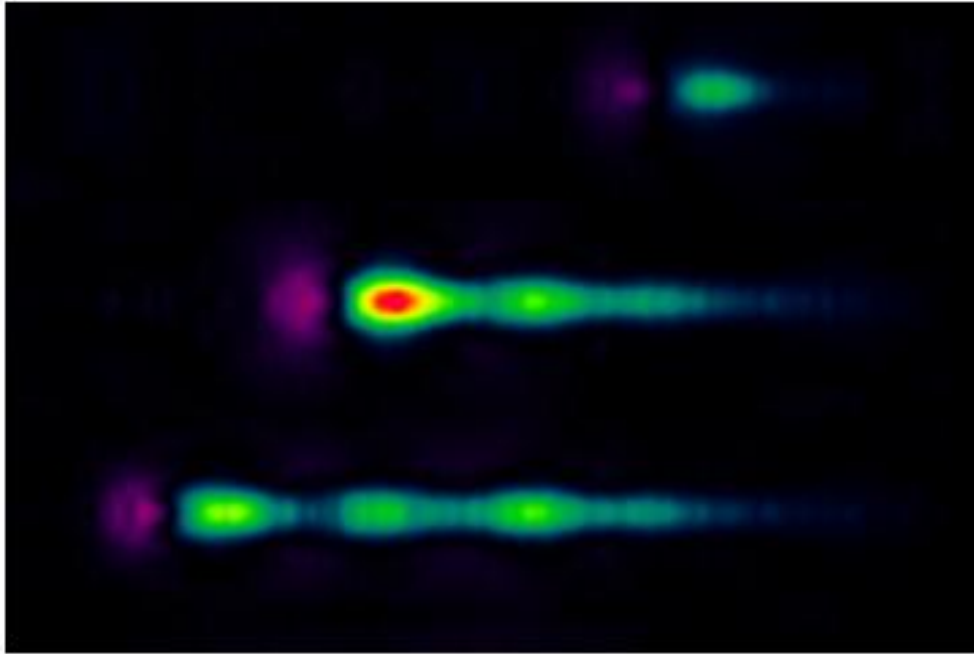
$$P(t) = -eNx(t) = \boxed{P_{\omega}^{(1)} e^{-i\omega t}} + \boxed{P_0^{(2)} + P_{\omega}^{(2)} e^{-i2\omega t}} + \dots + cc$$

Anharmonic potential



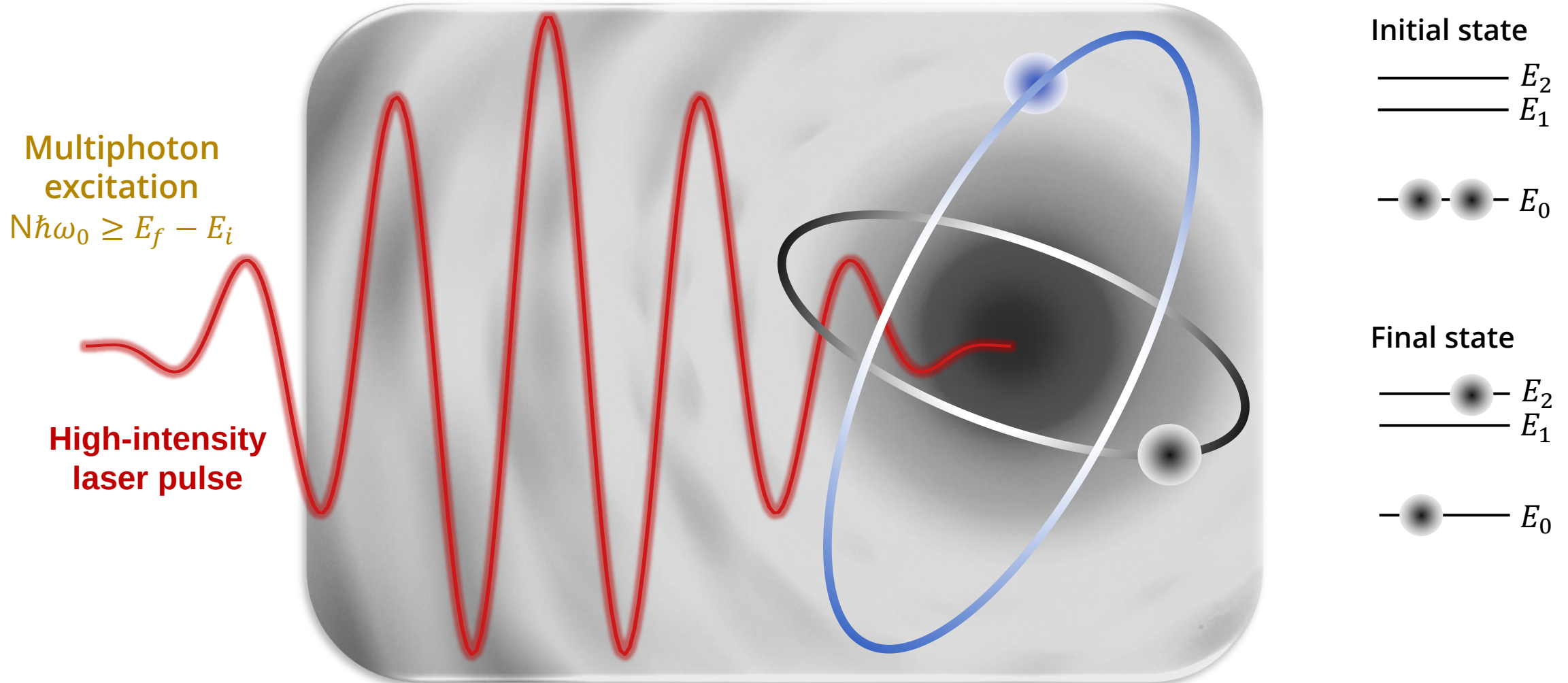
Examples of nonlinear optics phenomena

- Frequency-mixing and harmonic generation
- Filamentation
- Multiple focusing



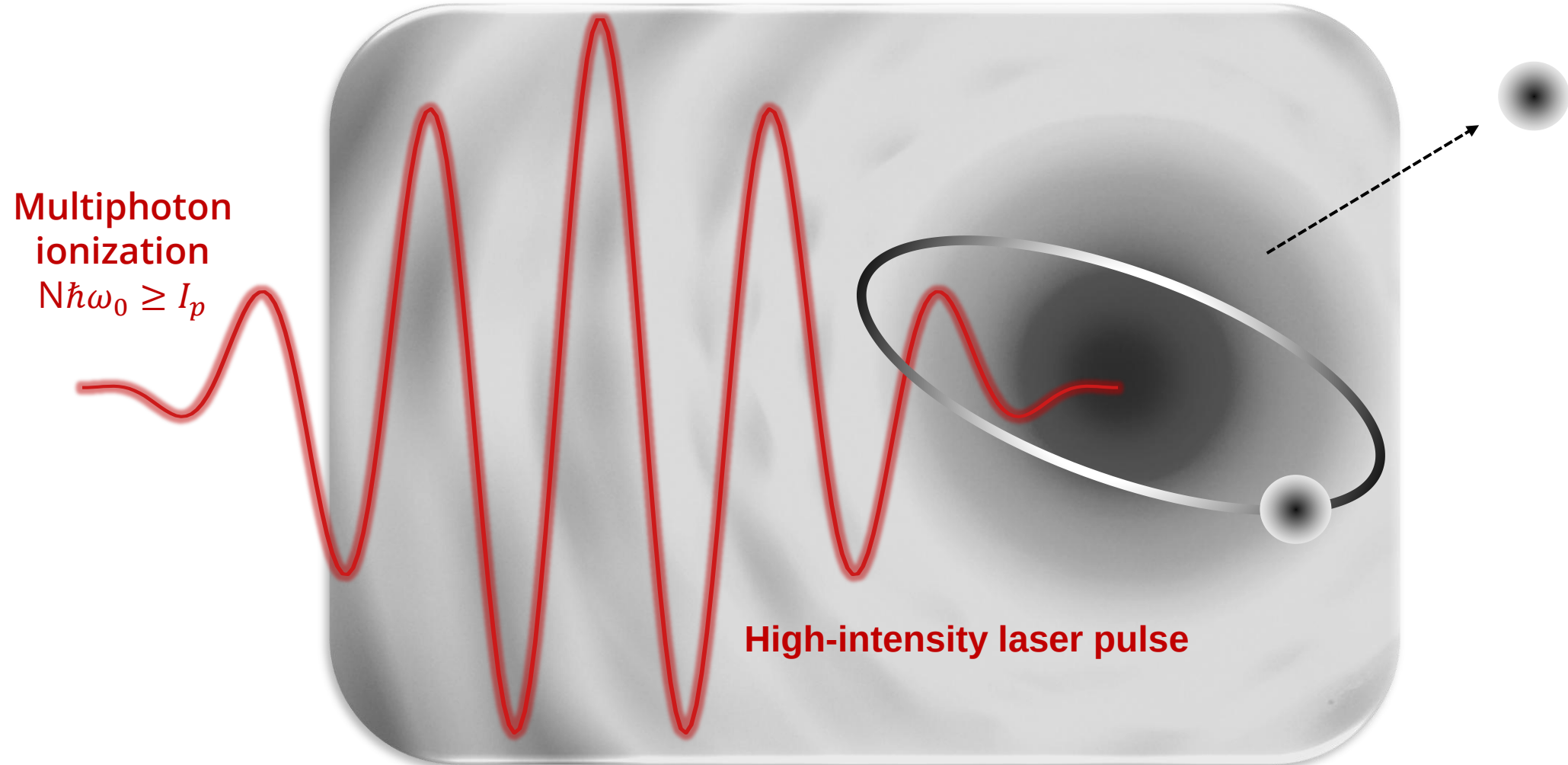
Nonlinear optical process: **Multiphoton excitation**

The quantum system absorbs the energy of several photons to reach a higher energy state during the interaction with a **high-intensity** laser pulse



Nonlinear optical process: **Multiphoton ionization**

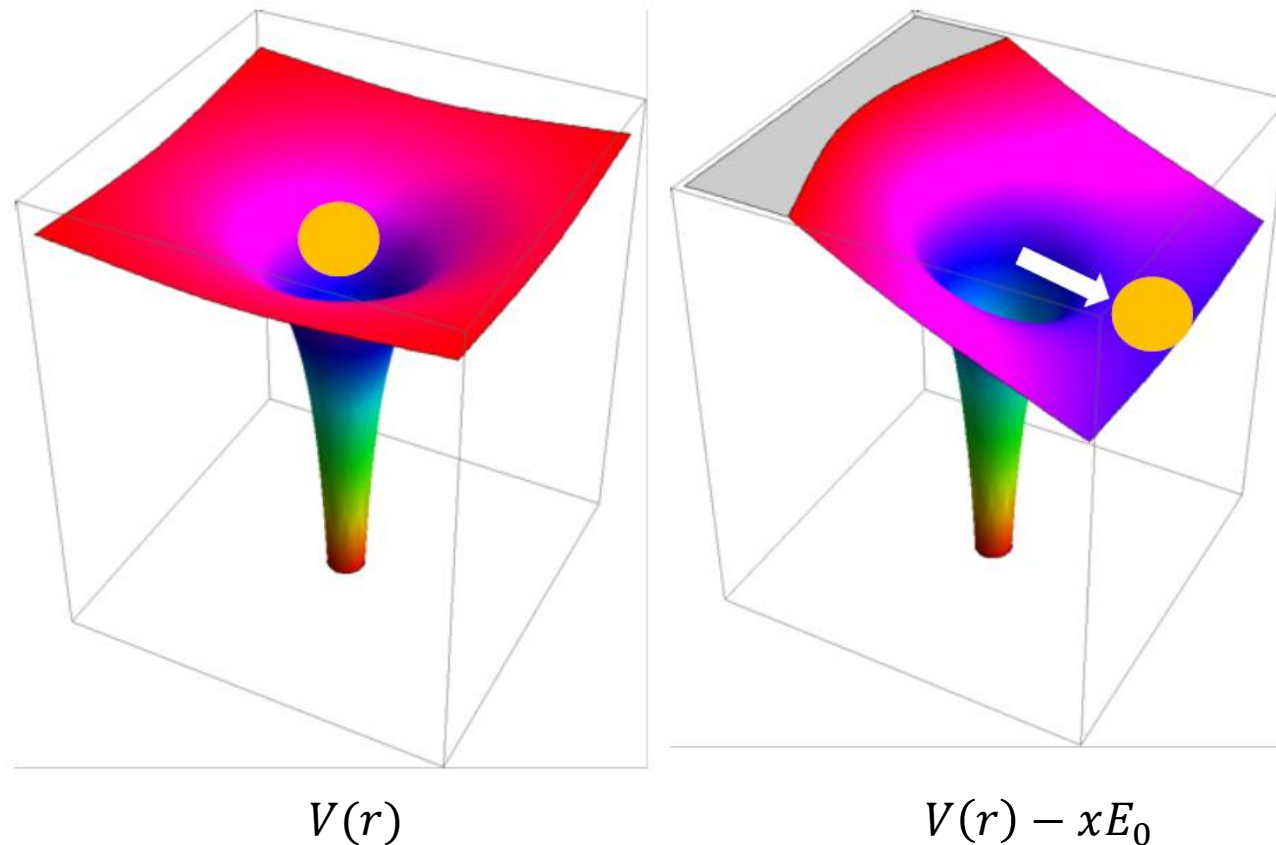
One or more electrons are released during the interaction with a **high intensity** laser pulse



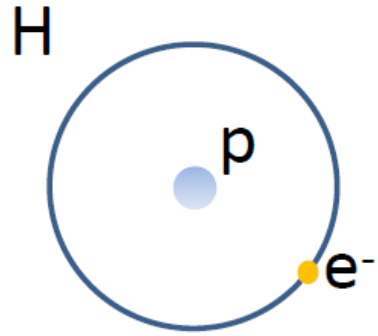
Fundamentals of Strong-Field Physics

Strong-Field Physics: beyond the limit of the perturbative approach

- The perturbative approach fails when the electric field is comparable to, or exceeds, the Coulombic potential binding valence electrons



Strong-Field Physics: beyond the limit of the perturbative approach



$$V(r) = -\frac{e^2}{4\pi\epsilon_0 r}$$

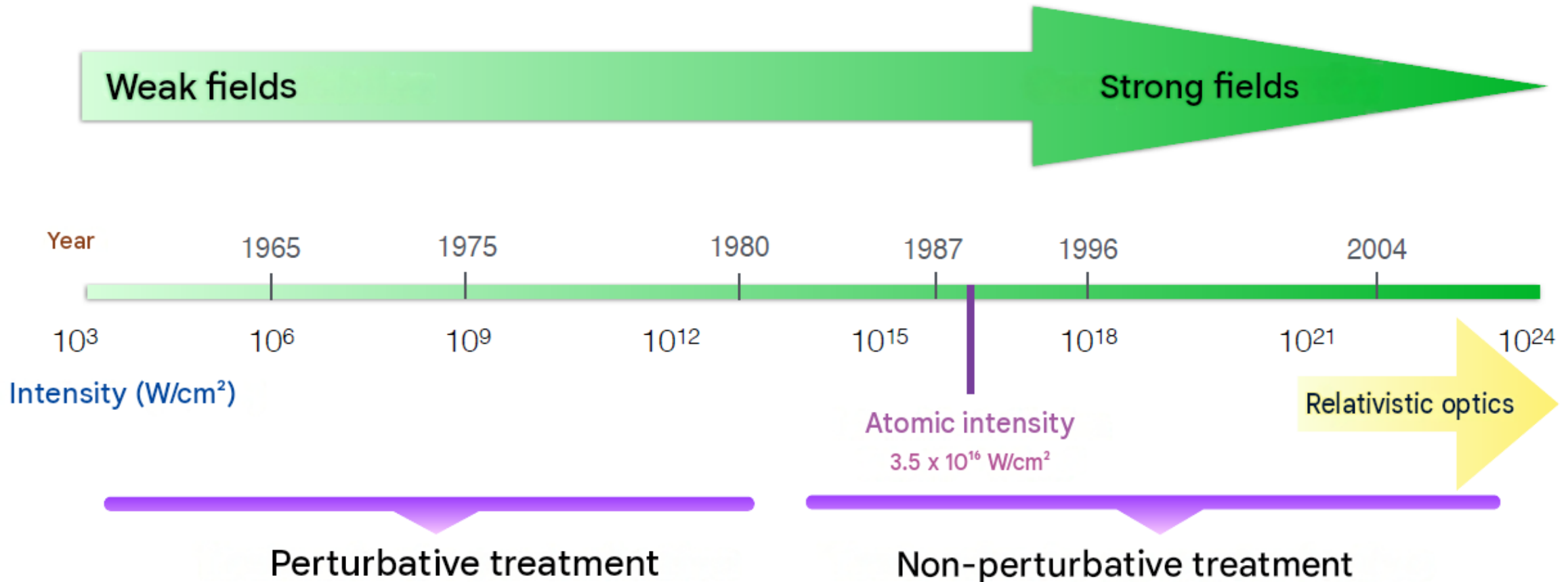
Bohr radius: $a_0 = 4\pi\epsilon_0 \frac{\hbar^2}{me^2}$

What electric field strength exerts a force equal to the Coulomb force?

$$F(r) = e \frac{\partial V(r)}{\partial r} = -\frac{e^2}{4\pi\epsilon_0 r^2} \quad \left| F(a_0) \right| = \frac{e^2}{4\pi\epsilon_0 a_0^2}$$

$$\left| F(a_0) \right| = eE_0 \quad \Rightarrow \quad \begin{aligned} E_0 &\approx 5.1 \times 10^{11} \text{ V} \cdot \text{m}^{-1} \\ I_0 &\approx 3.5 \times 10^{16} \text{ W} \cdot \text{cm}^{-2} \end{aligned}$$

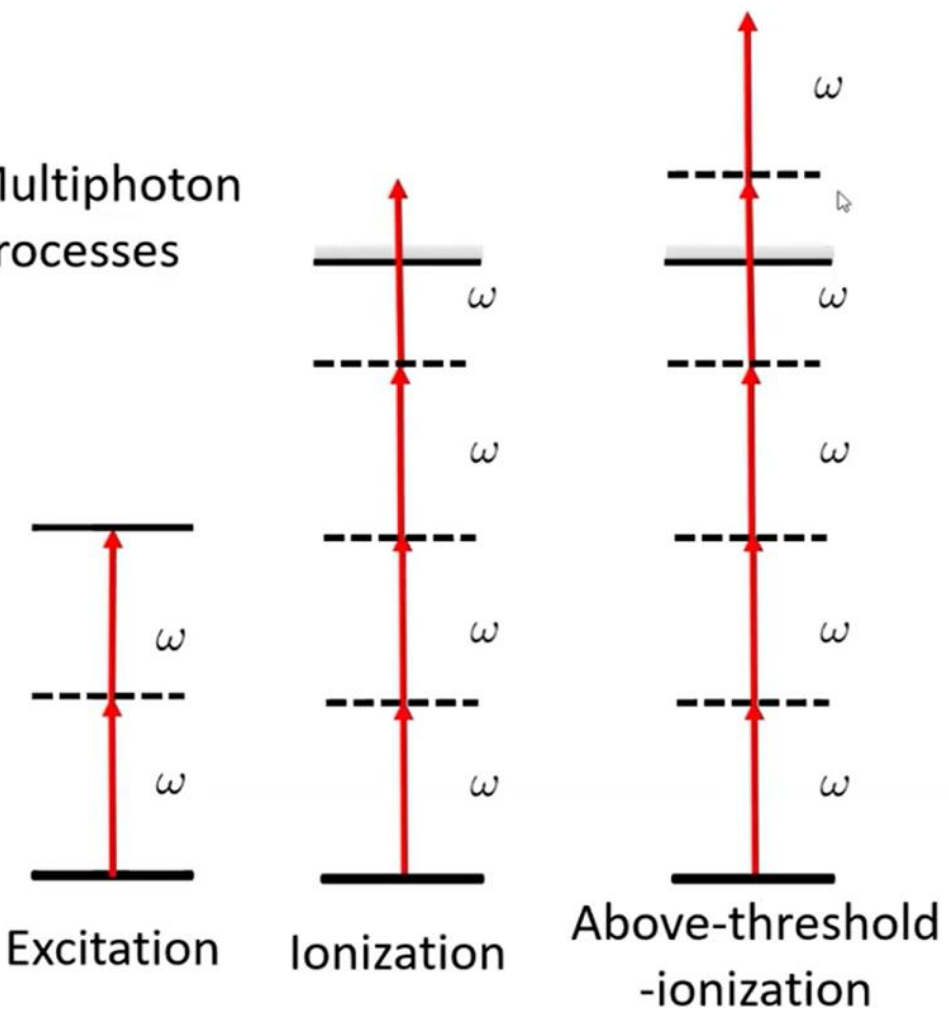
Strong-Field Physics: beyond the limit of the perturbative approach



Multiphoton processes

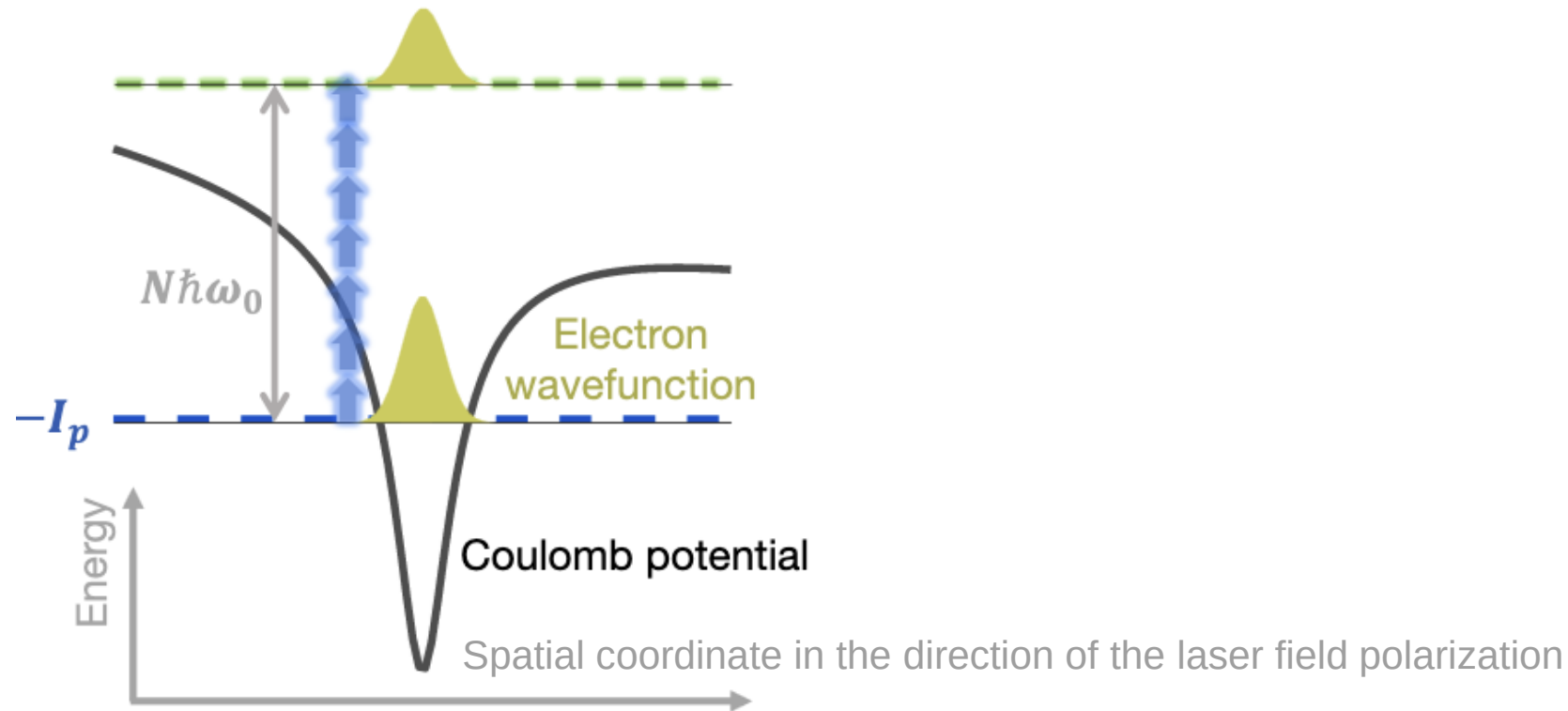


Multiphoton processes

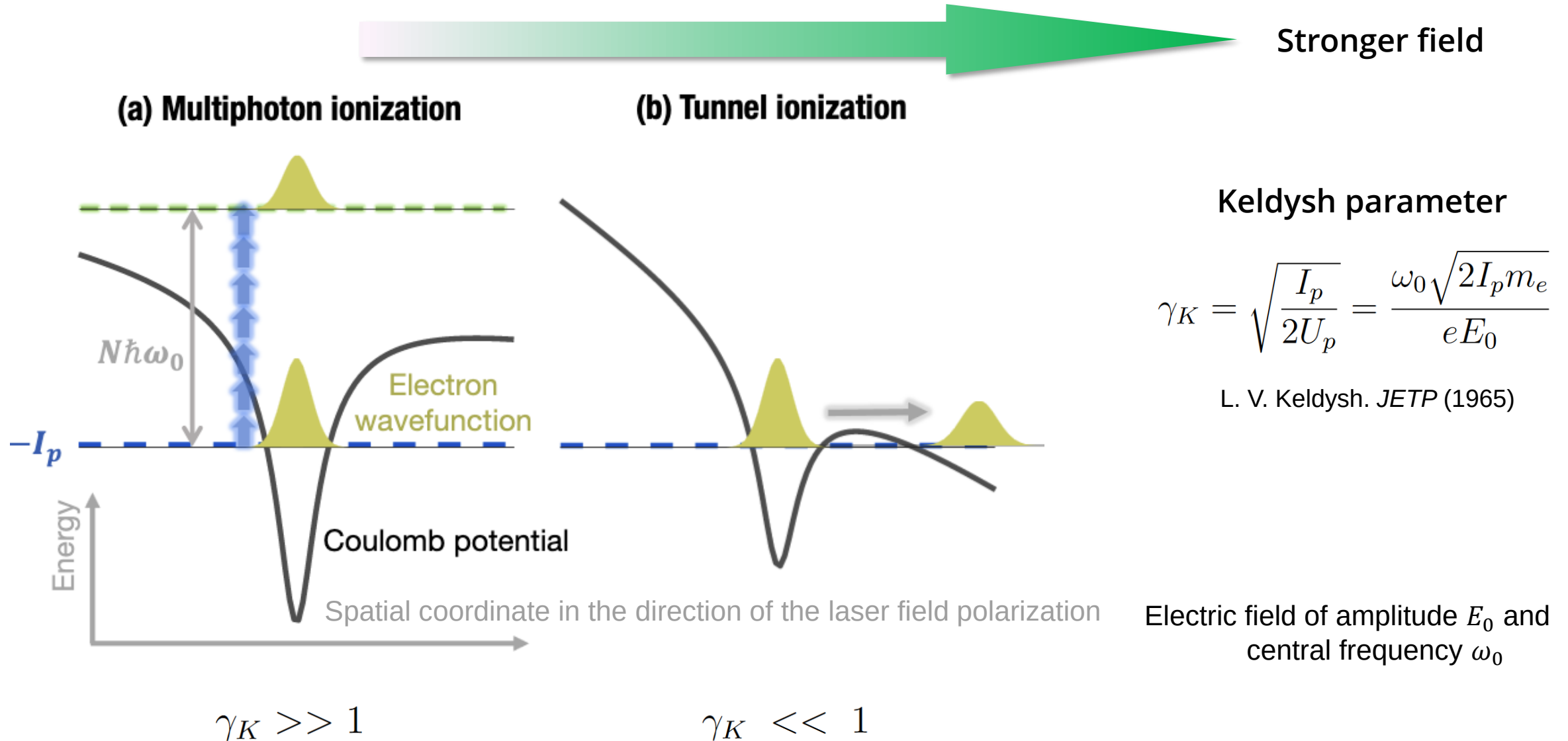


Strong-field ionization regimes in laser-atom interactions

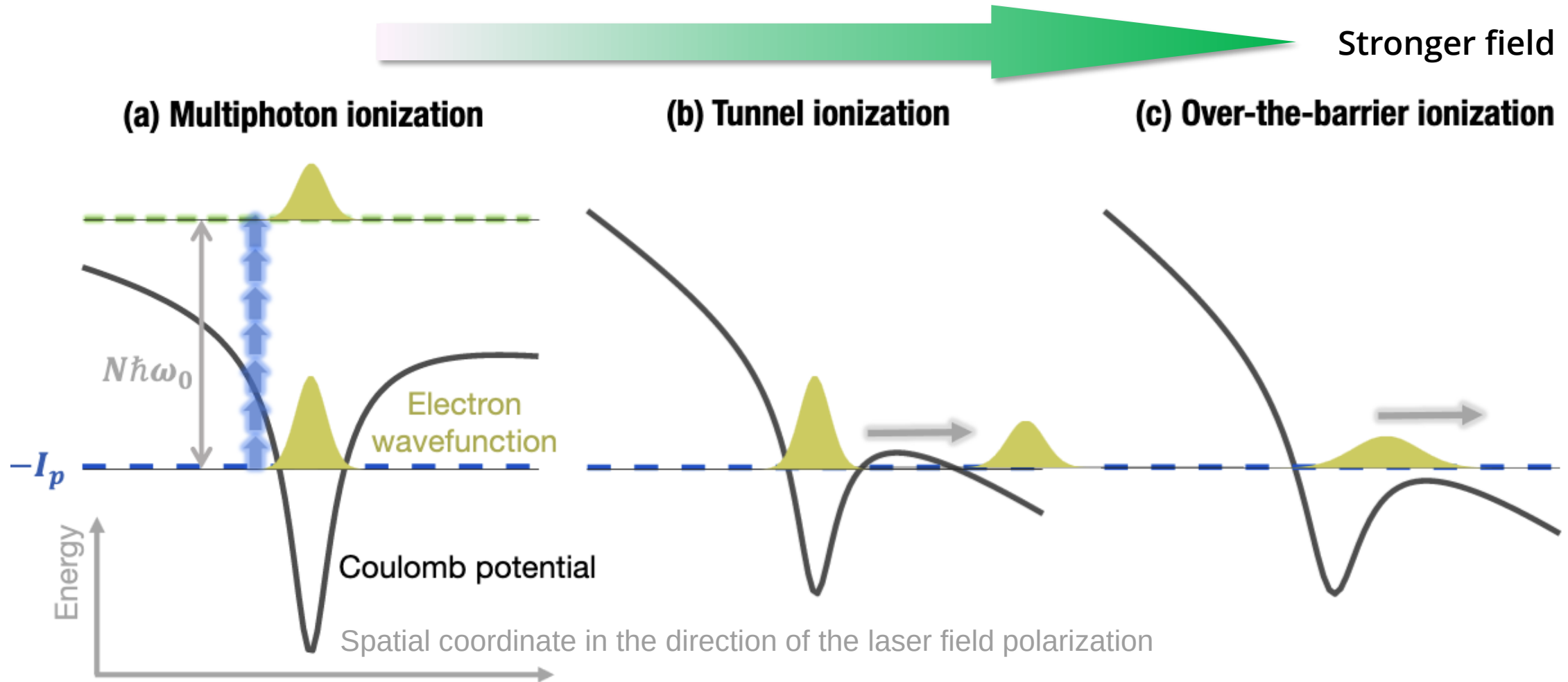
(a) Multiphoton ionization



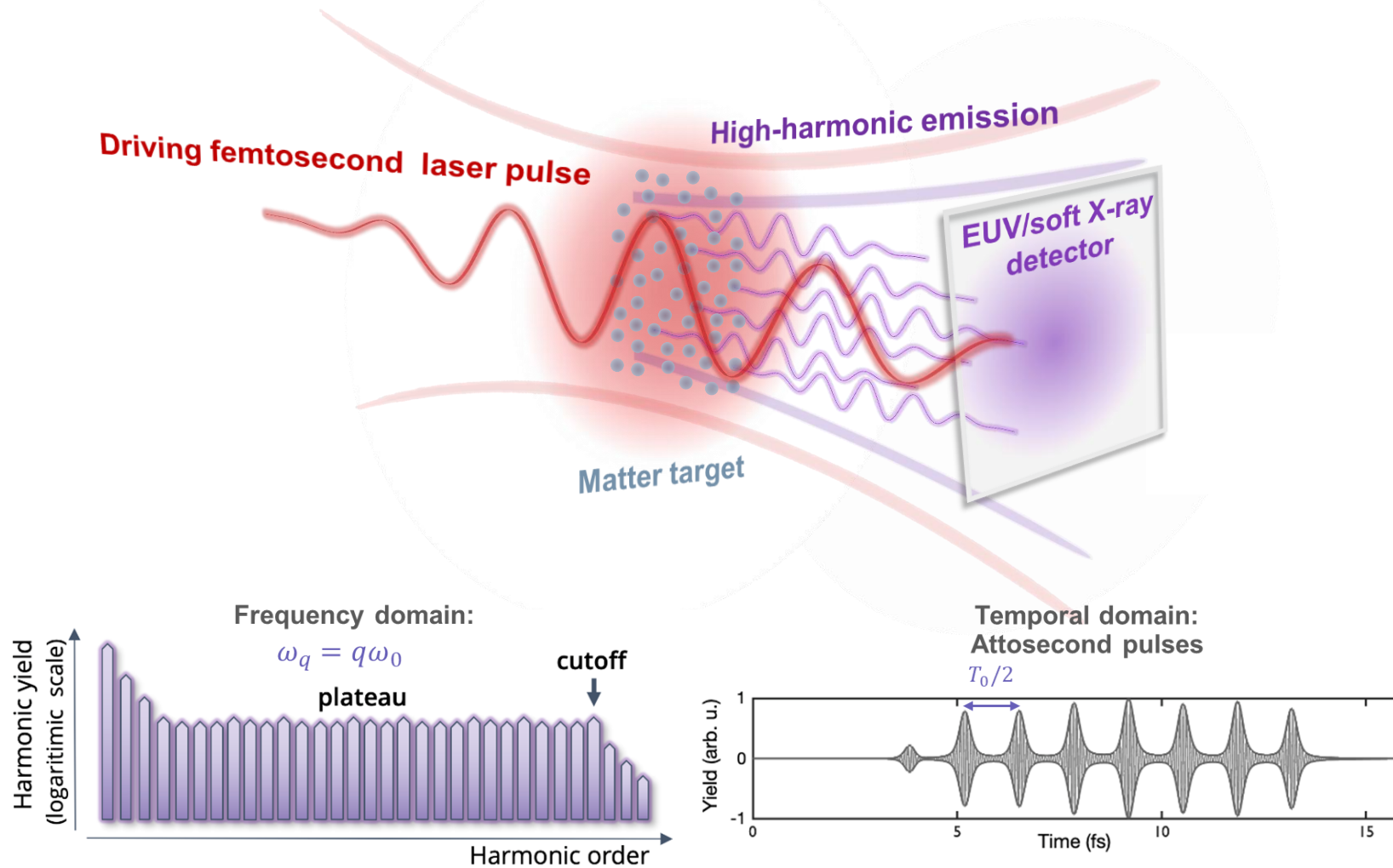
Strong-field ionization regimes in laser-atom interactions



Strong-field ionization regimes in laser-atom interactions

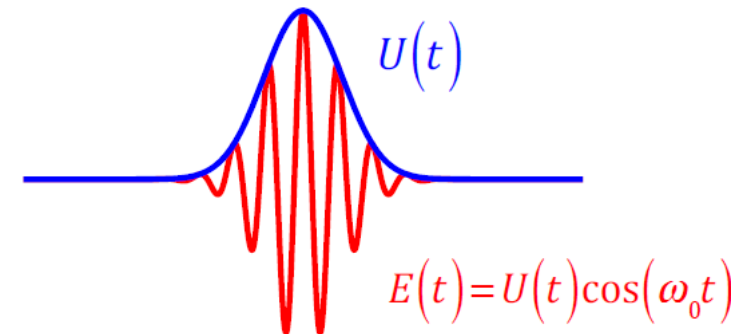
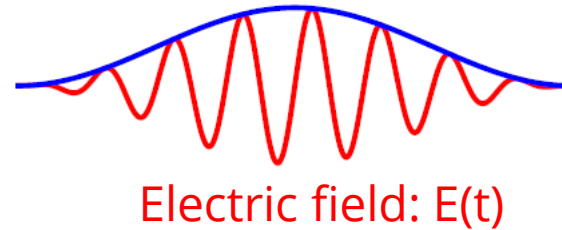
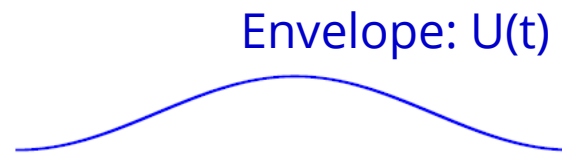
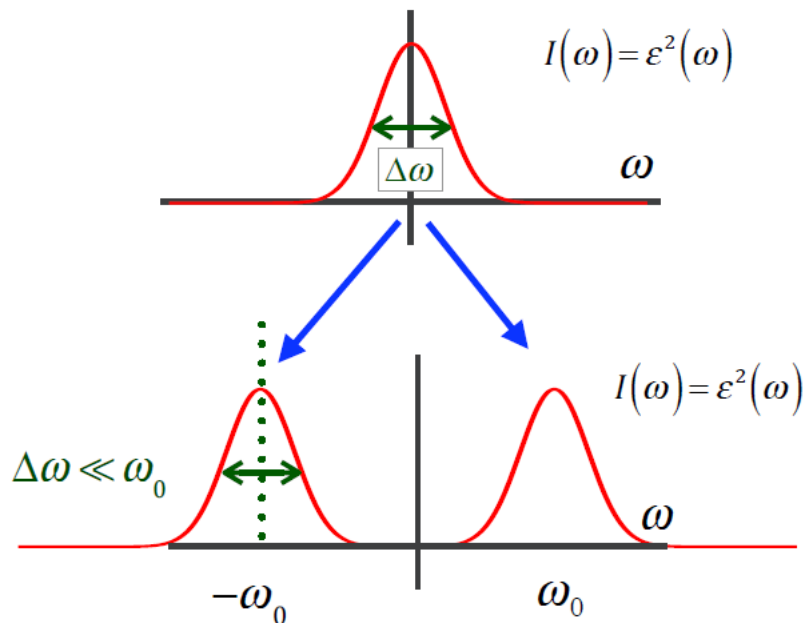


High-order harmonic generation (HHG)

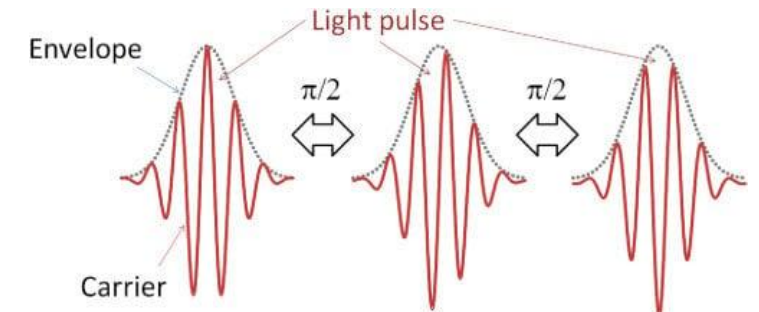


Fourier transform to switch from frequency to time domains

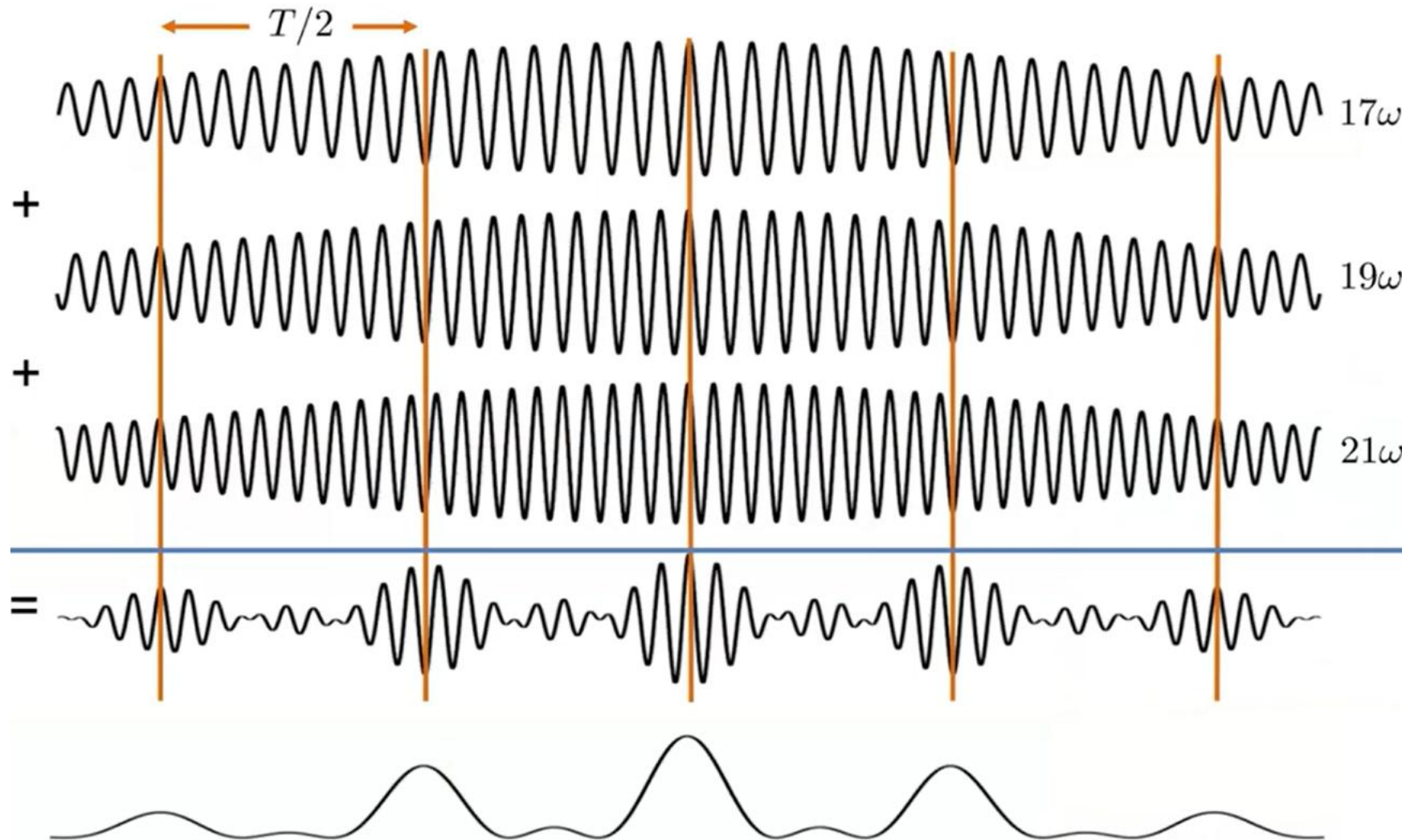
$$E(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{E}(\omega) e^{-i\omega t} d\omega \quad \longleftrightarrow \quad \tilde{E}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} E(t) e^{+i\omega t} dt$$



$$E(t) = U(t) \cos(\omega_0 t + \varphi_{CEP})$$

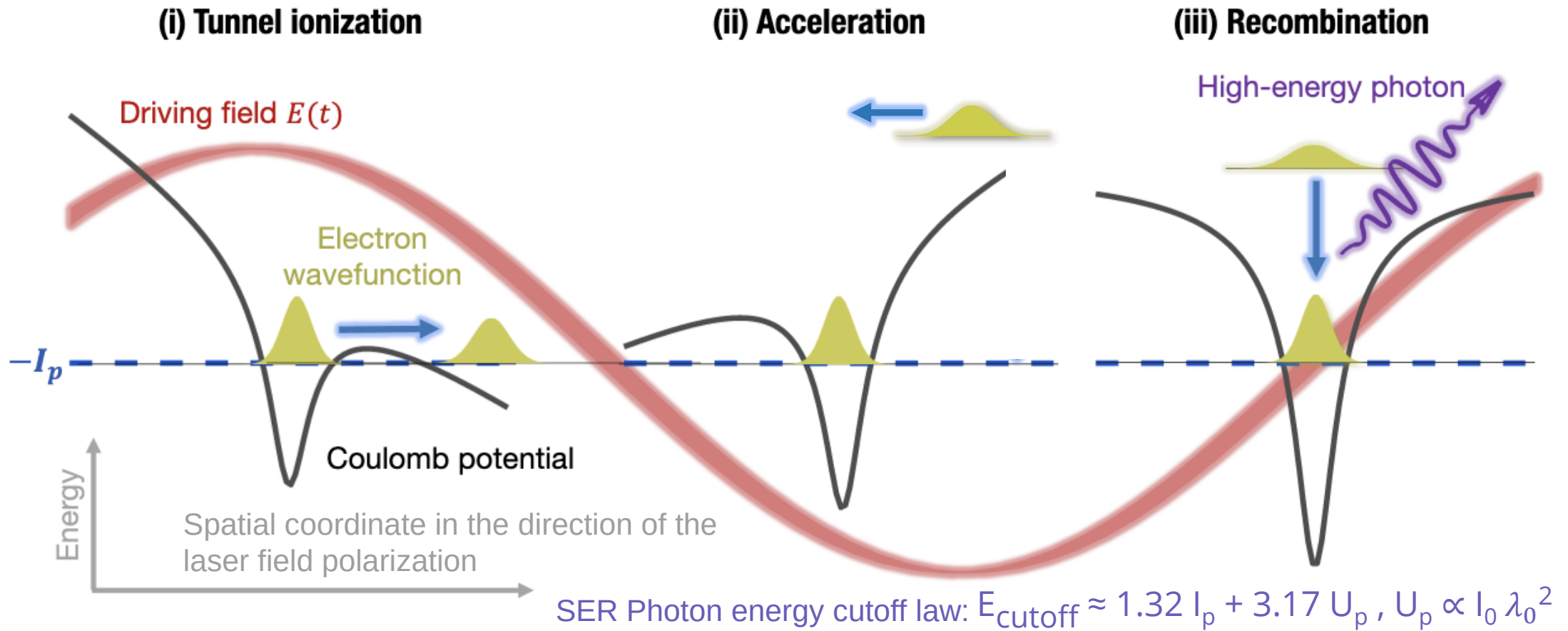


Attosecond pulse train from a coherent superposition of harmonics



High-order harmonic generation (HHG) at the atomic level

Three-step process



K. J. Schafer et al. *Phys. Rev. Lett.* (1993)
P. B. Corkum. *Phys. Rev. Lett.* (1993); M. Lewenstein et al. *Phys. Rev. A* (1994)

J. Krause et al. *Phys. Rev. Lett.* (1992)
M. Lewenstein et al. *Phys. Rev. A* (1994)

Brief historical perspective about HHG

1987-1989 First experiments of HHG and identification of the high harmonic plateau

A. McPherson *et al.* *JOSA B* (1987); M. Ferray *et al.* *J. Phys. B* (1988); X. F. Li *et al.* *Phys. Rev. A* (1989)

1991-1994 Important theoretical advances in the understanding of HHG

A. L'Huillier *et al.* *J. Phys. B* (1991); J. Krause *et al.* *Phys. Rev. Lett.* (1992); G. Farkas & C. Tóth. *Phys. Lett. A* (1992); K. J. Schafer *et al.* *Phys. Rev. Lett.* (1993); P. B. Corkum. *Phys. Rev. Lett.* (1993); M. Lewenstein *et al.* *Phys. Rev. A* (1994)

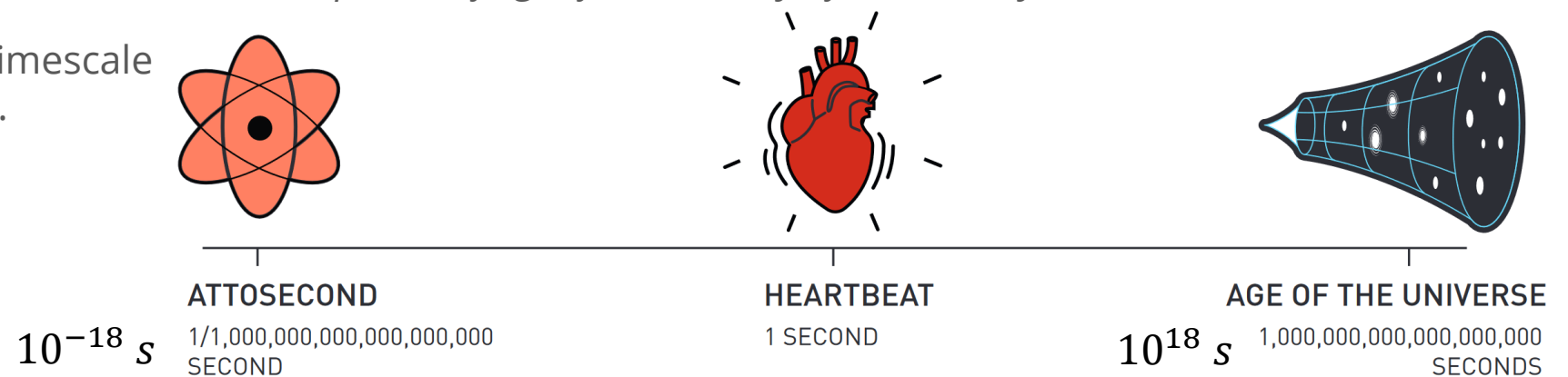
2001 Experimental characterization of attosecond pulses

P. M. Paul *et al.* *Science* (2001); M. Hentschel *et al.* *Nature* (2001)

2023 Nobel Prize in Physics to Pierre Agostini, Ferenc Krausz, and Anne L'Huillier

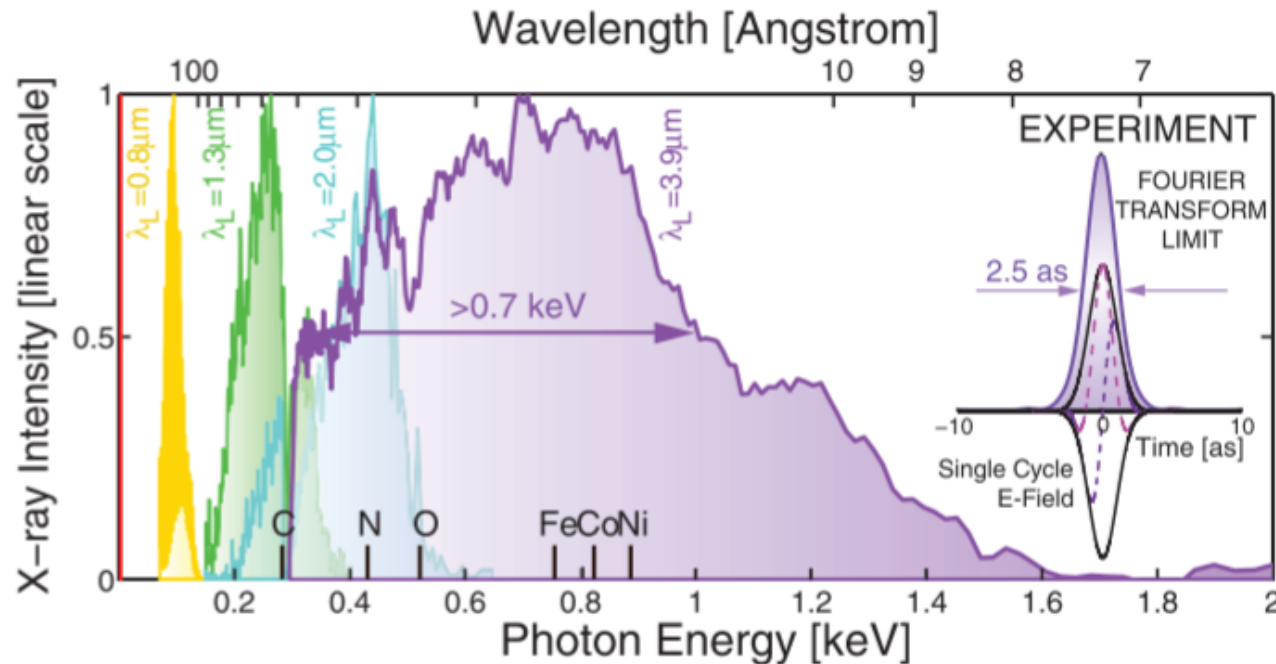
“for experimental methods that generate attosecond pulses of light for the study of electron dynamics in matter”

The attosecond is the natural timescale of electron dynamics in matter.



Actual records in the experiments of attosecond light sources

Broader spectrum: 0.7 keV

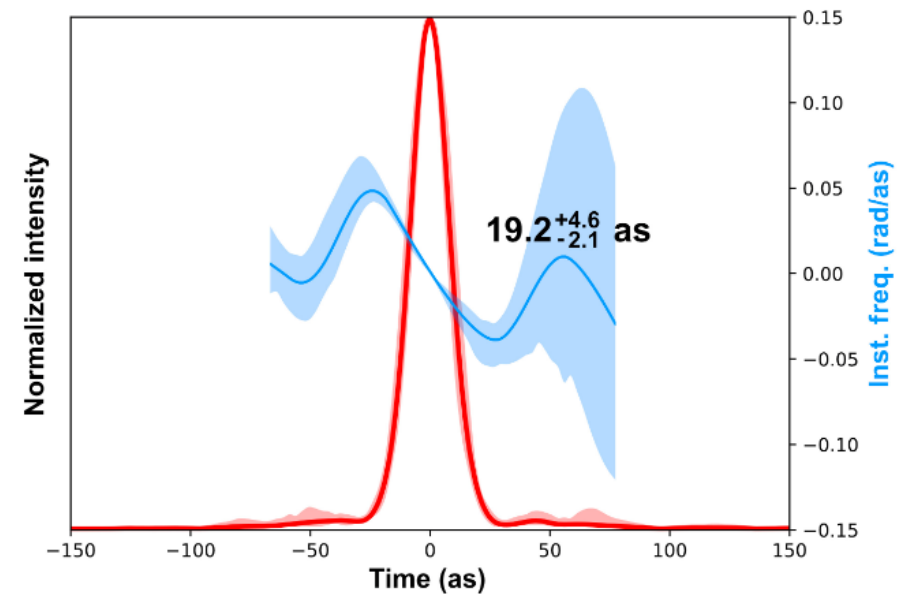


harmonic 5000!

T. Popmintchev *et al.* *Science* (2012)

Shortest laser pulse: 19.2 as

Shorter than the
atomic unit of time (24.2 as)!



F. Ardana-Lamas *et al.* *Ultrafast Science* (2025)

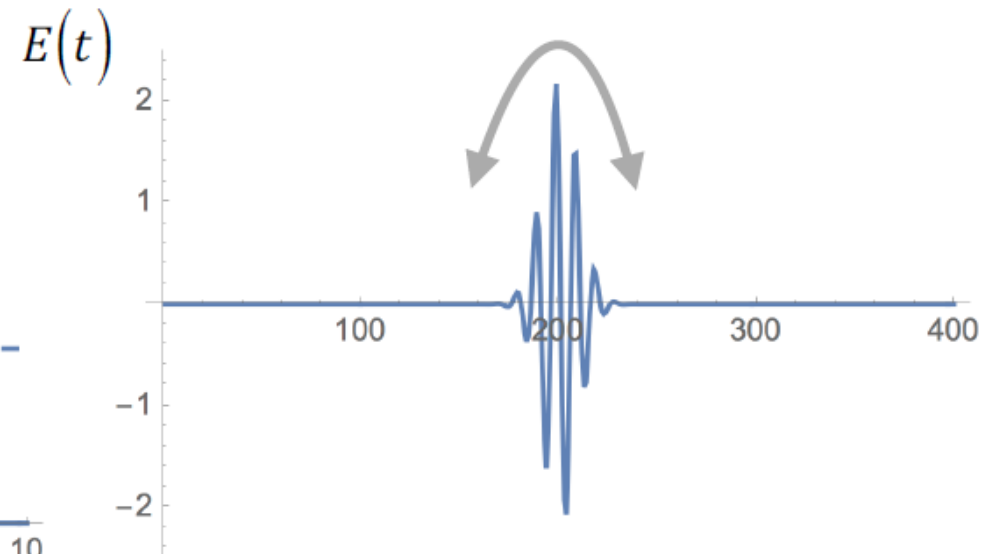
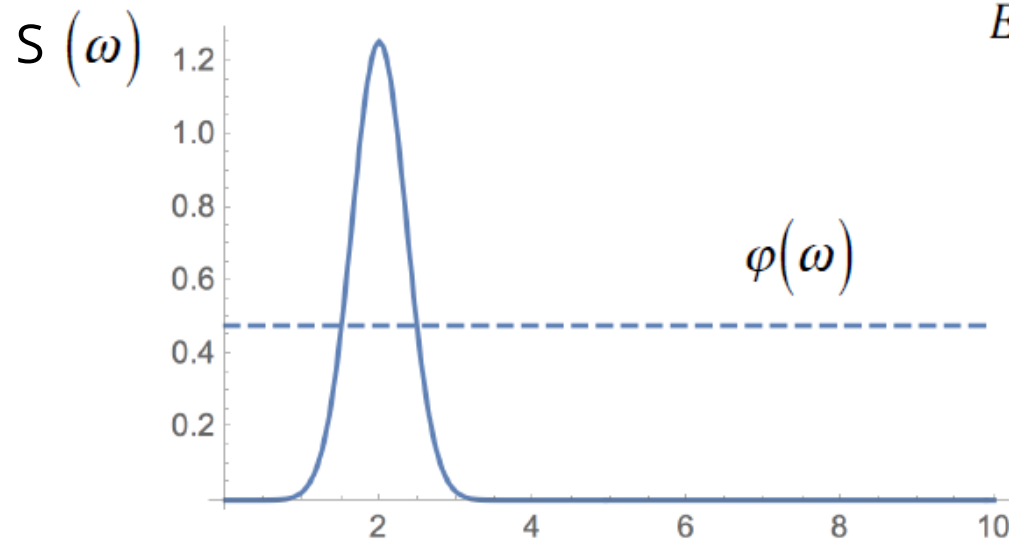
$$\Delta E \Delta t \geq C$$

Role of the spectral phase

- A broad spectrum does not necessarily imply a short pulse duration!

$$\varphi(\omega) = a_0 + a_1\omega + a_2\omega^2 + a_3\omega^3$$

a_0 $\Updownarrow$ Constant phase (CEP)

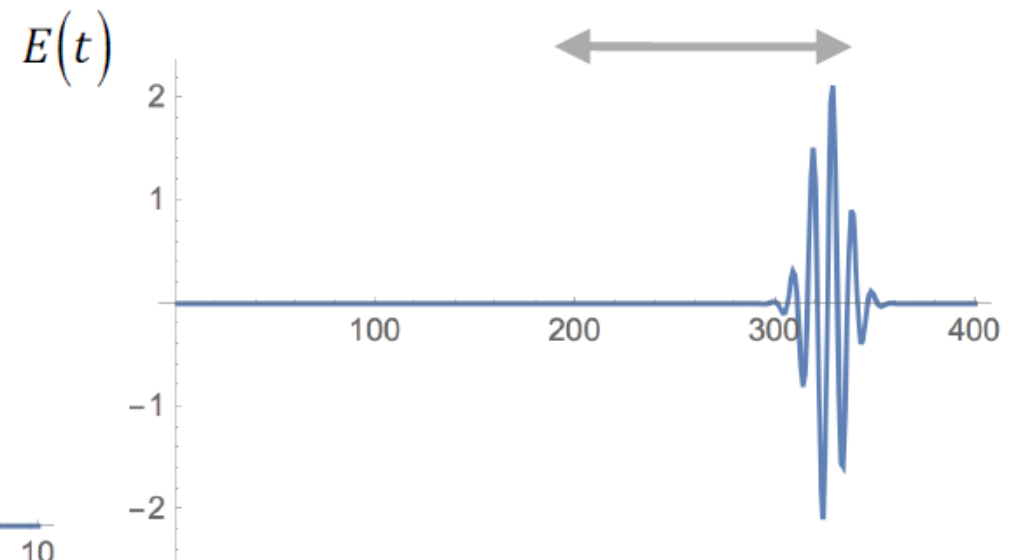
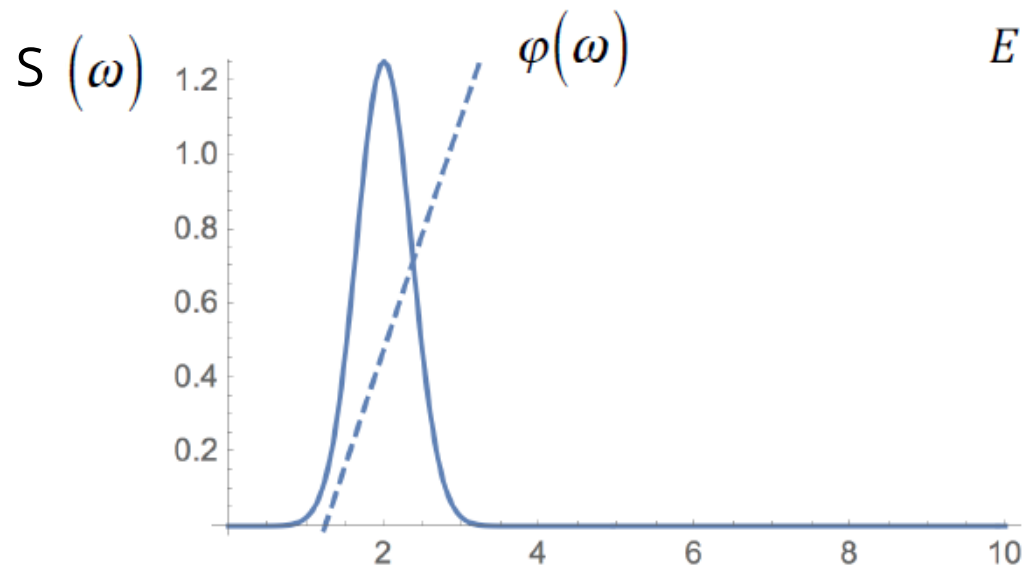


Role of the spectral phase

- A broad spectrum does not necessarily imply a short pulse duration!

$$\varphi(\omega) = a_0 + a_1\omega + a_2\omega^2 + a_3\omega^3$$

a_1 $\updownarrow$ Linear phase

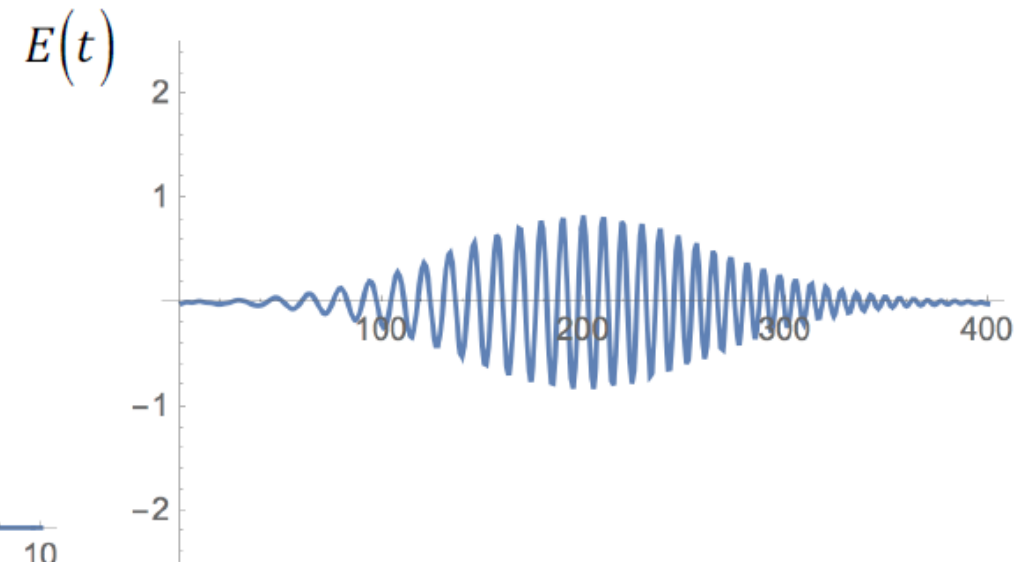
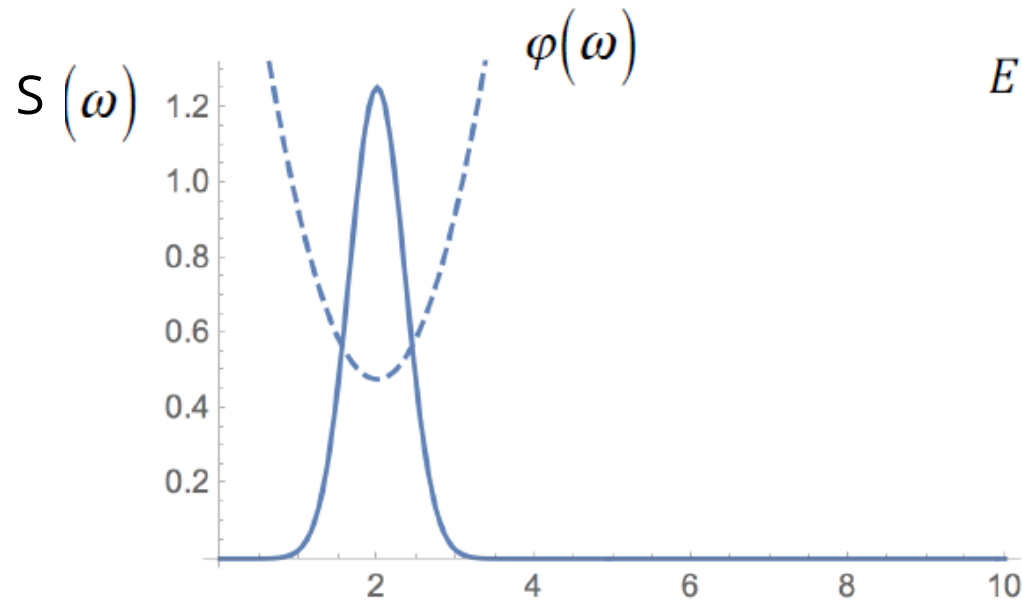


Role of the spectral phase

- A broad spectrum does not necessarily imply a short pulse duration!

$$\varphi(\omega) = a_0 + a_1\omega + a_2\omega^2 + a_3\omega^3$$

$a_2 \updownarrow$ Quadratic phase (linear chirp)

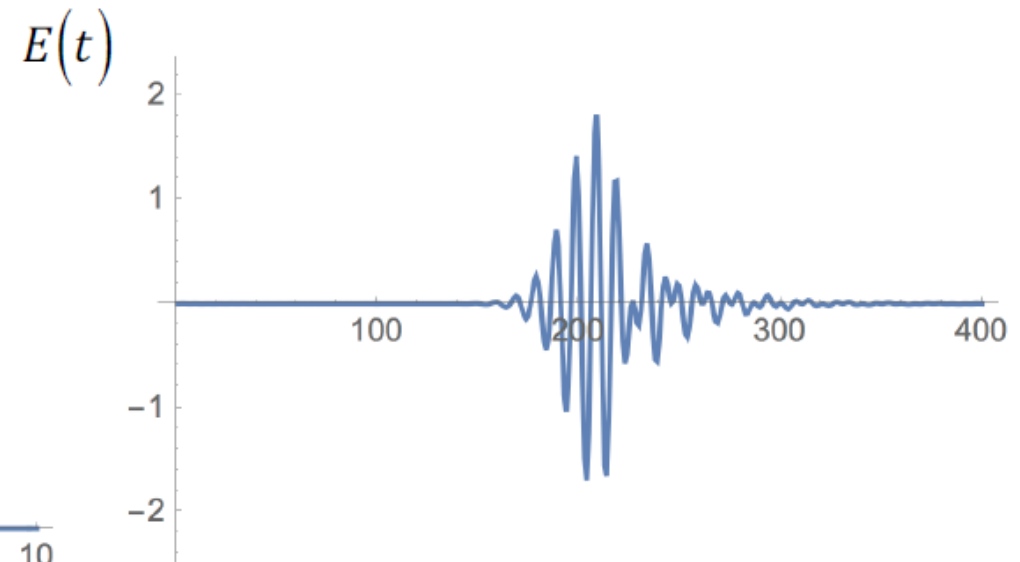
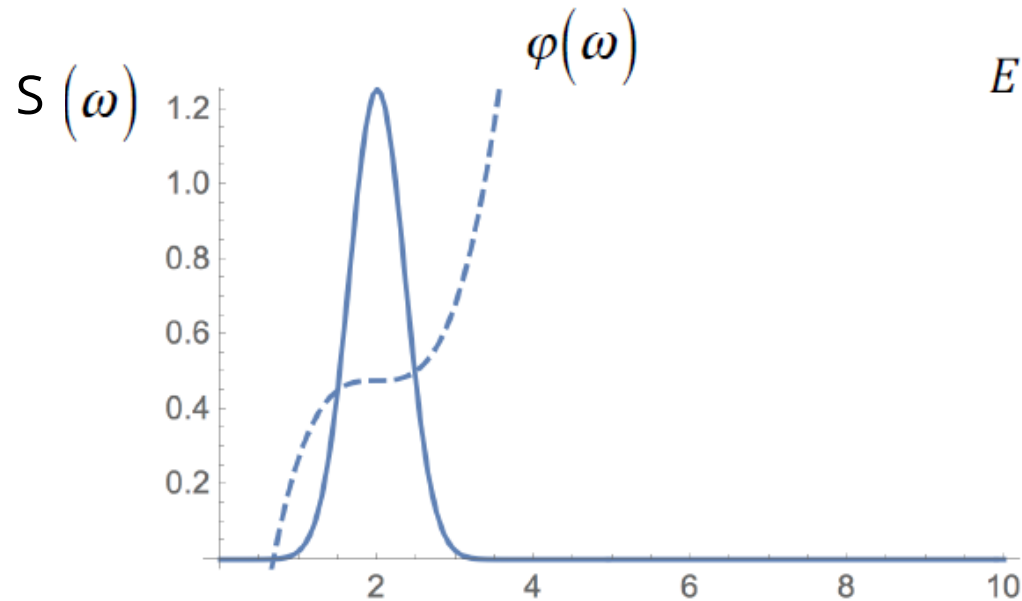


Role of the spectral phase

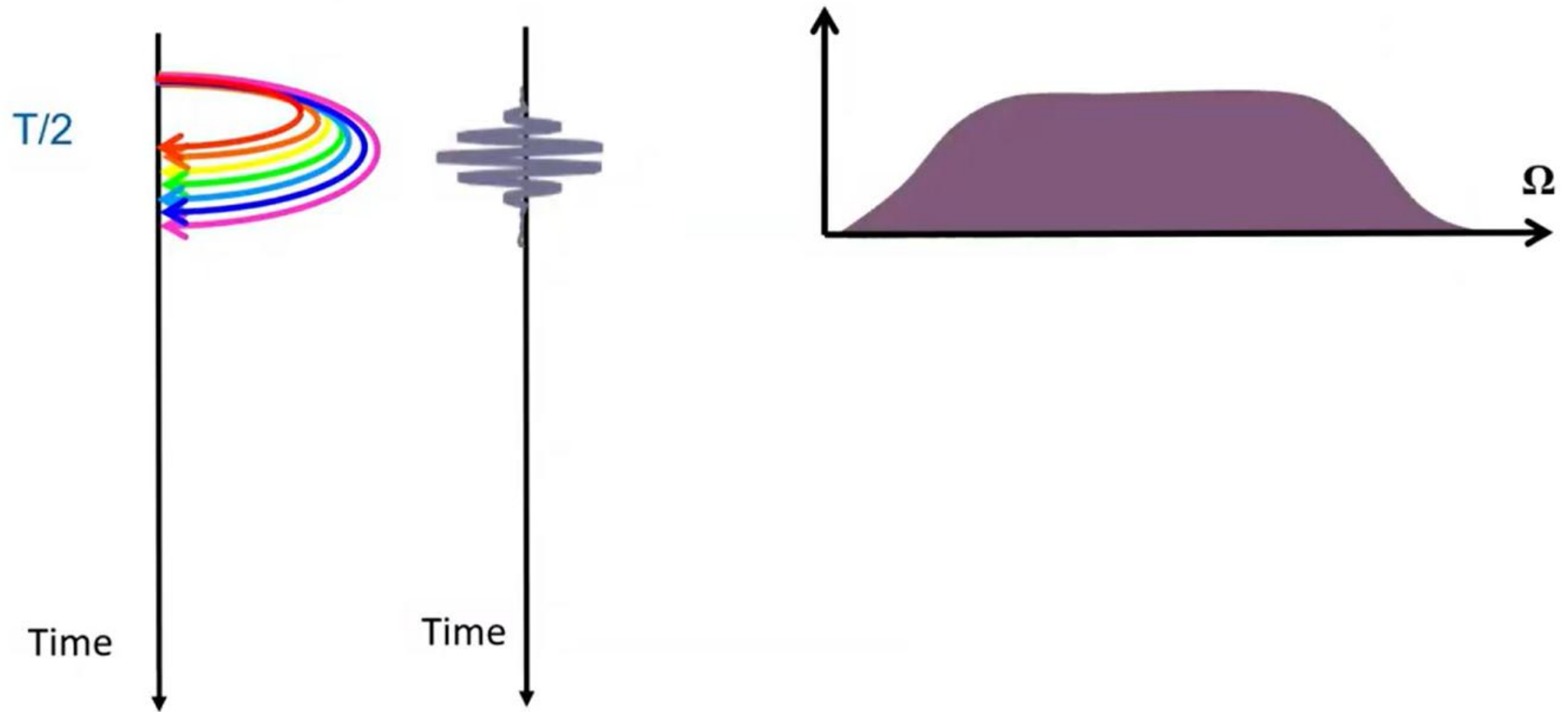
- A broad spectrum does not necessarily imply a short pulse duration!

$$\varphi(\omega) = a_0 + a_1\omega + a_2\omega^2 + a_3\omega^3$$

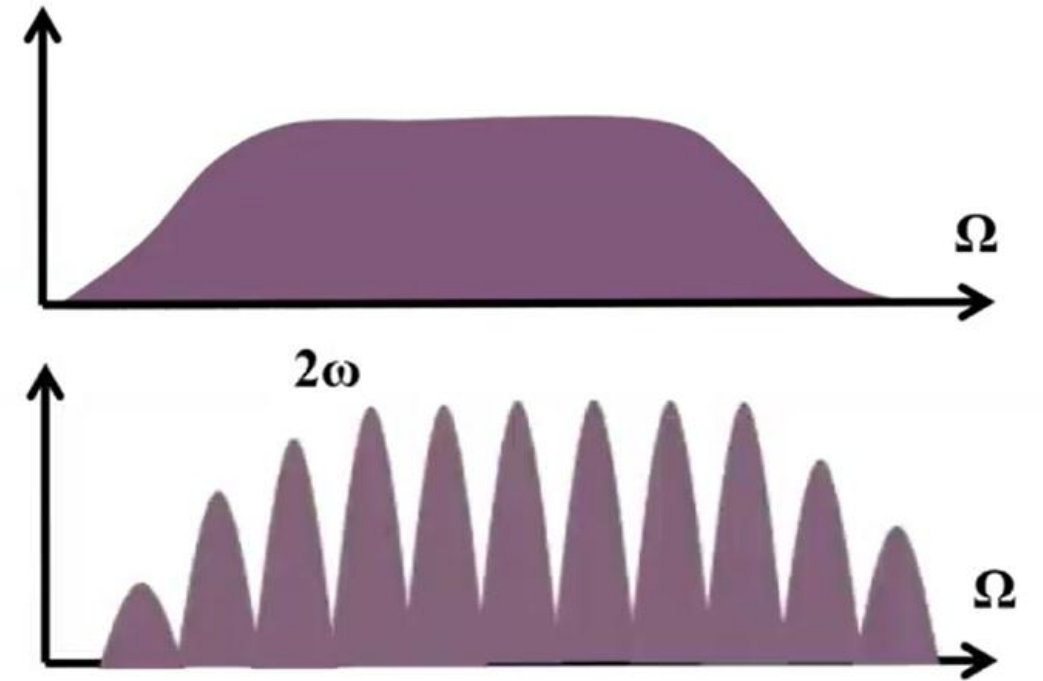
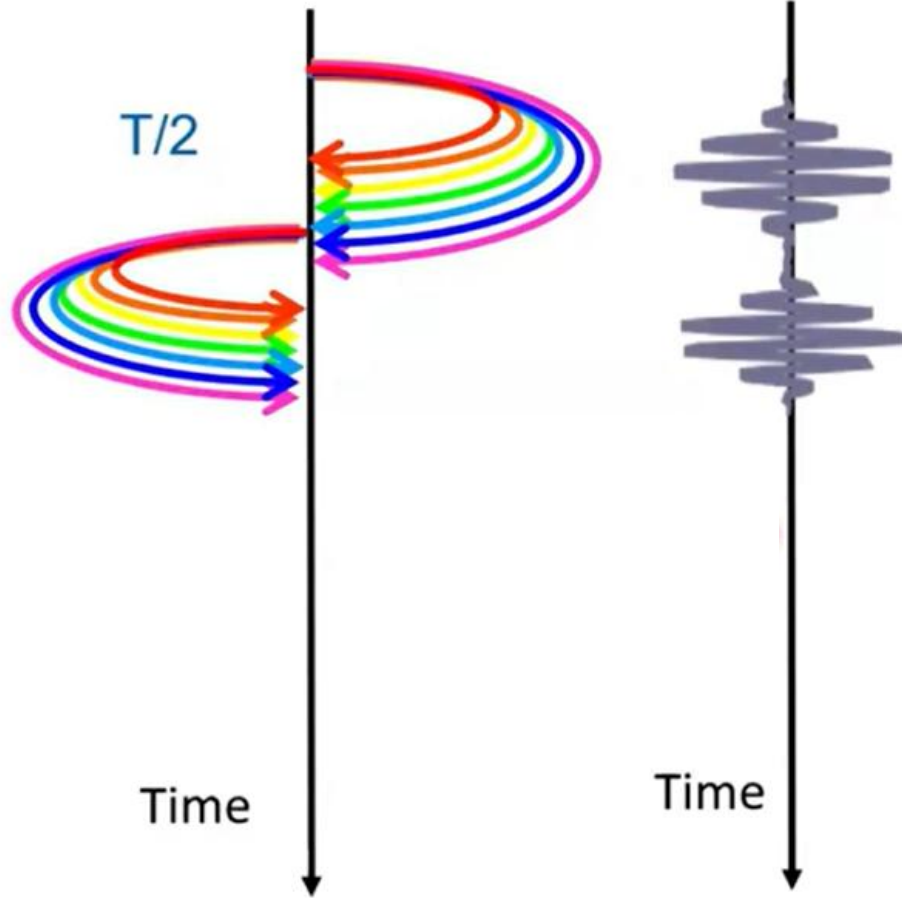
$a_3 \updownarrow$ Cubic phase (nonlinear chirp)



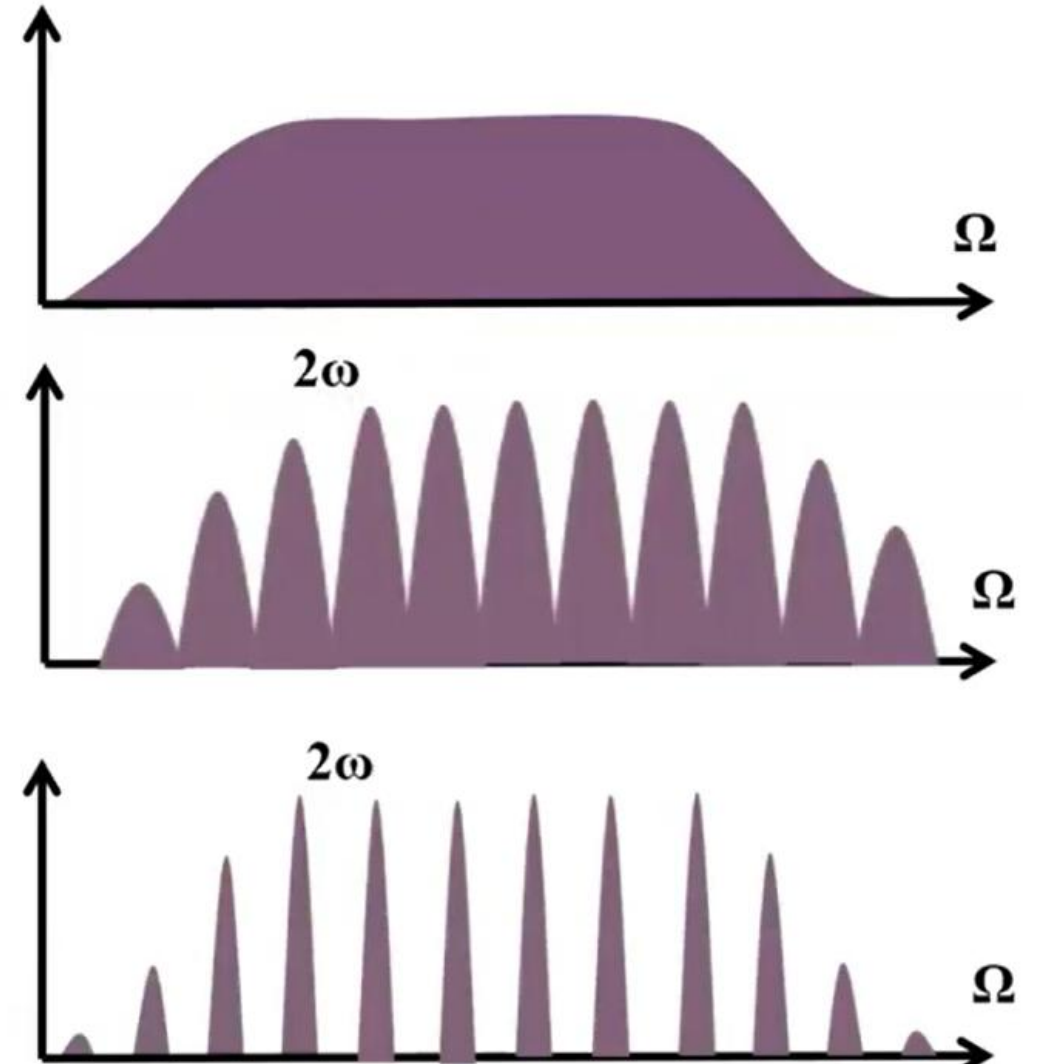
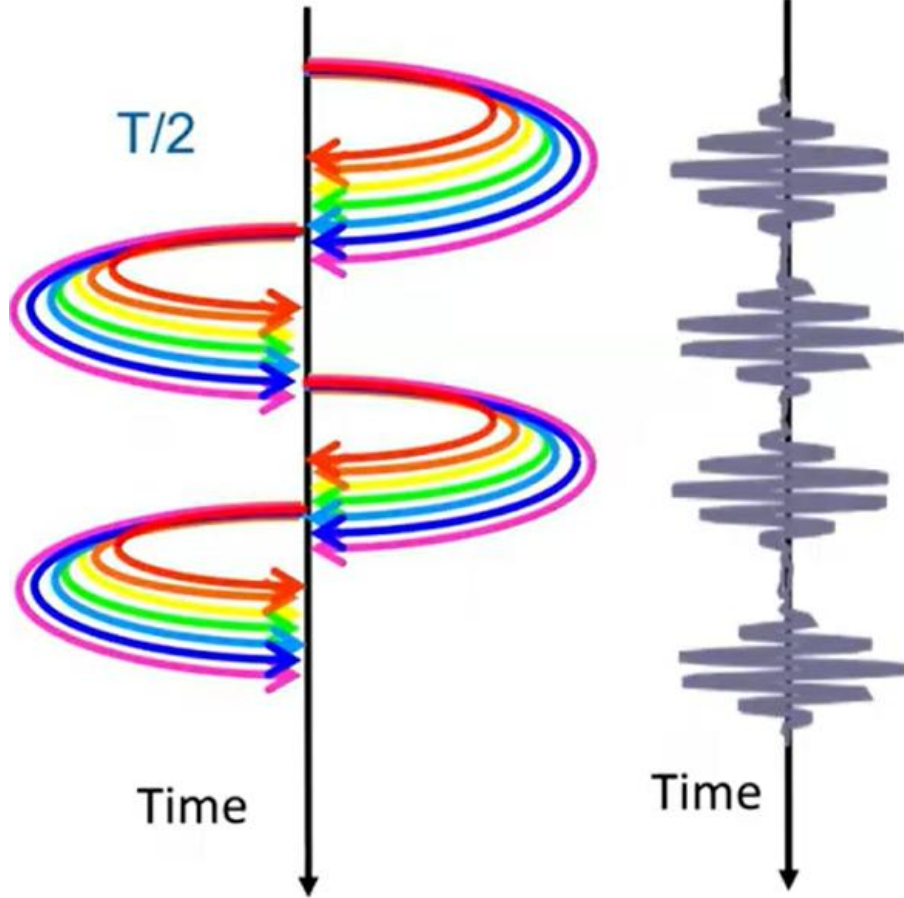
Supercontinuum spectra vs sharper harmonic peaks



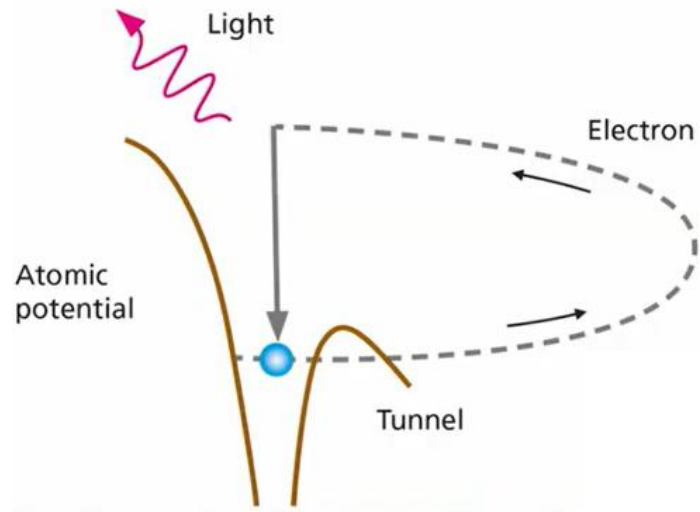
Supercontinuum spectra vs sharper harmonic peaks



Supercontinuum spectra vs sharper harmonic peaks



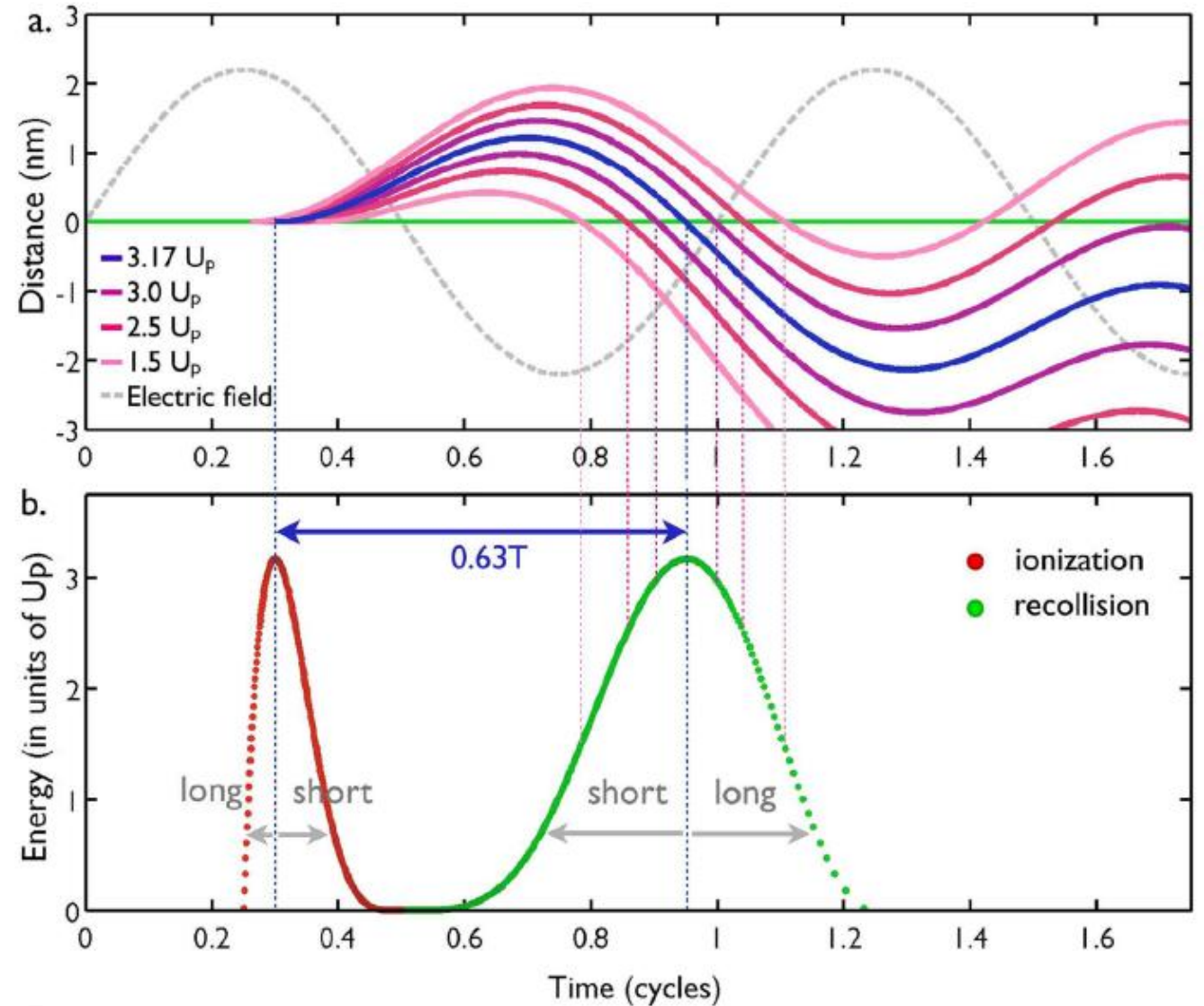
Classical free-electron trajectories



$$\mathbf{a}(t) = -\frac{e}{m_e} \mathbf{E}(t)$$

$$\mathbf{v}(t) = -\frac{e}{m_e} \int_{t_0}^t \mathbf{E}(t') dt'$$

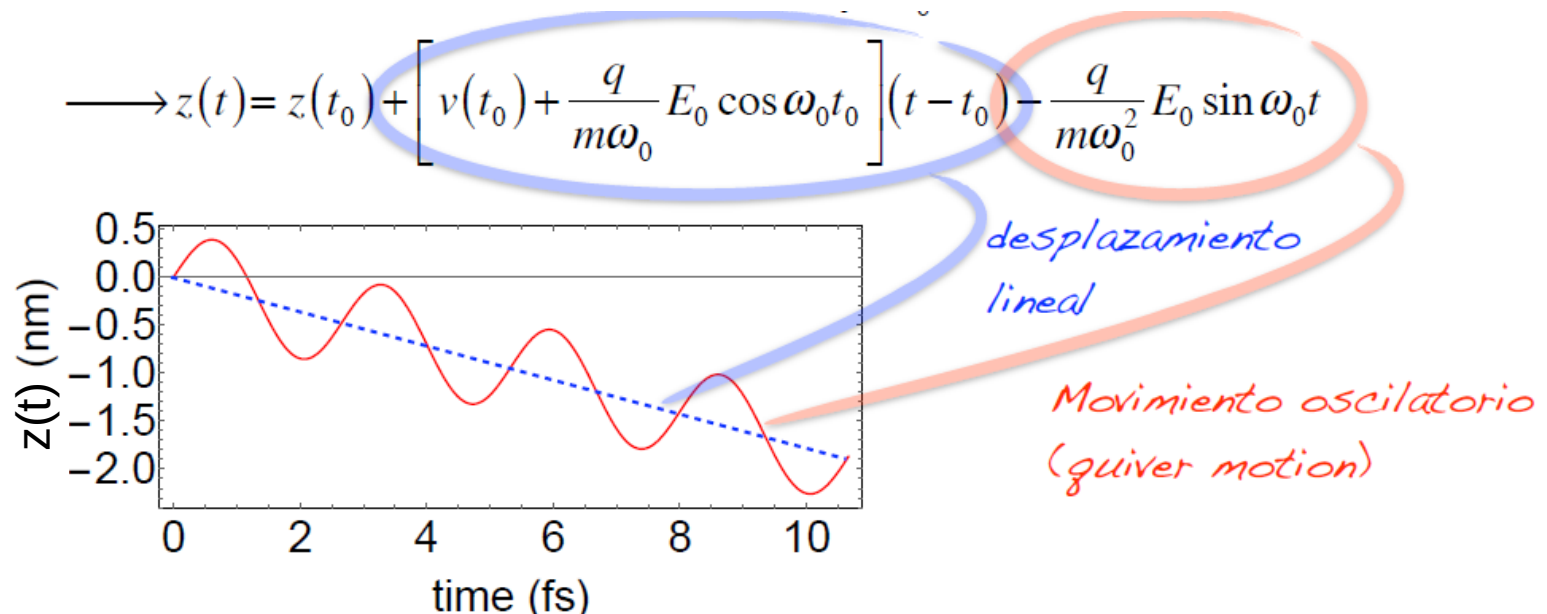
$$\mathbf{r}(t) = -\frac{e}{m_e} \int_{t_0}^t \left(\int_{t_0}^{t'} \mathbf{E}(t'') dt'' \right) dt'$$



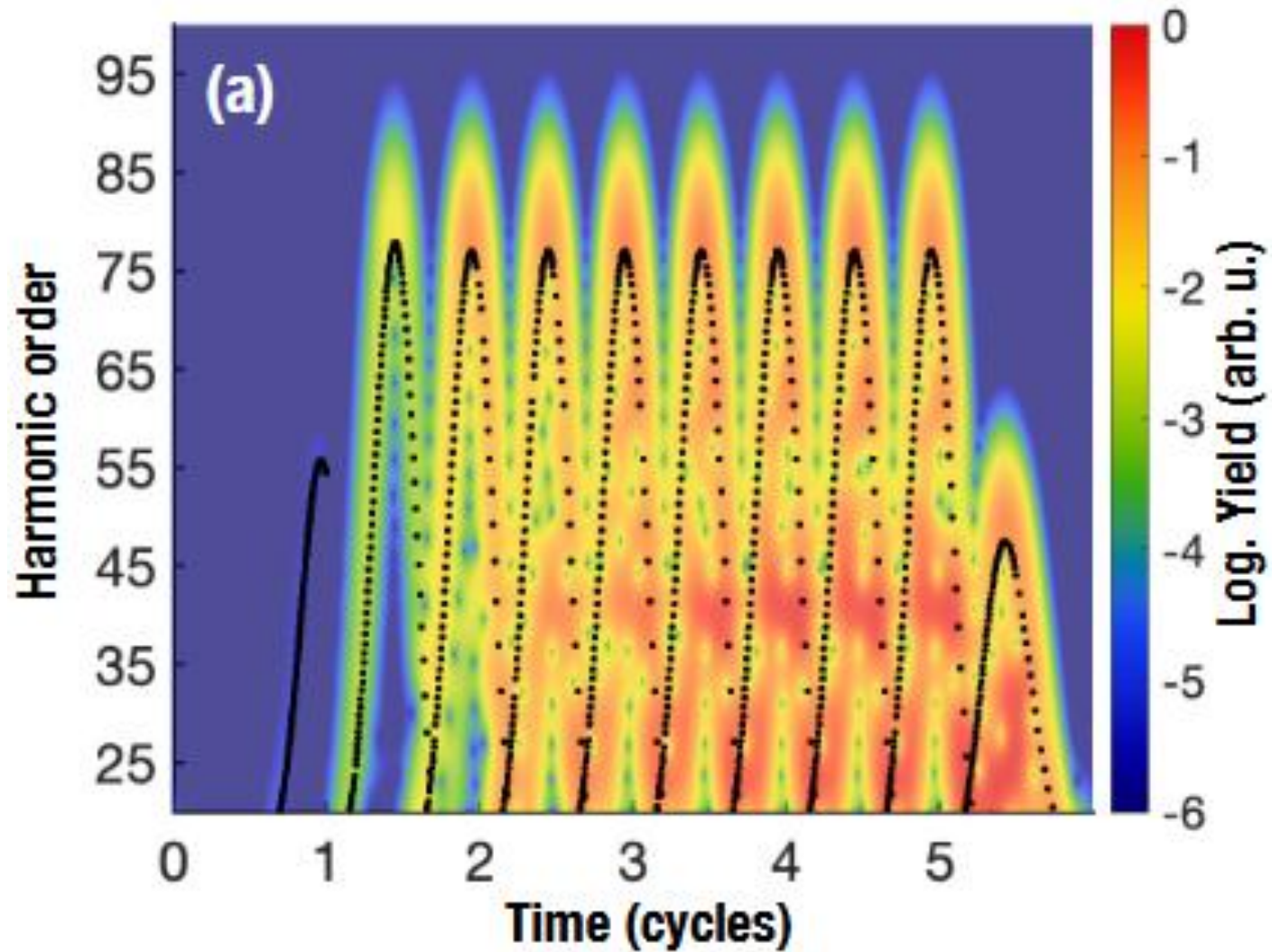
Classical free-electron trajectories

Important concepts:

- The ponderomotive energy is the average kinetic energy of a free electron interacting with a linearly polarized monochromatic electric field $U_p = \frac{e^2 E_0^2}{4m_e \omega_0^2}$
- The velocity of a free electron is given by the vector potential $\vec{v}(t) - \vec{v}(t_0) = -\frac{q}{mc} [\vec{A}(t) - A(t_0)]$
- The trajectory of a monochromatic field is characterized by a quiver motion and a linear displacement



Classical free-electron trajectories vs TDSE calculation



High harmonic emission towards the X-ray regime

The cutoff energy law: $E_c \approx I_p + 3.17 U_p$, $U_p \propto I \lambda^2$

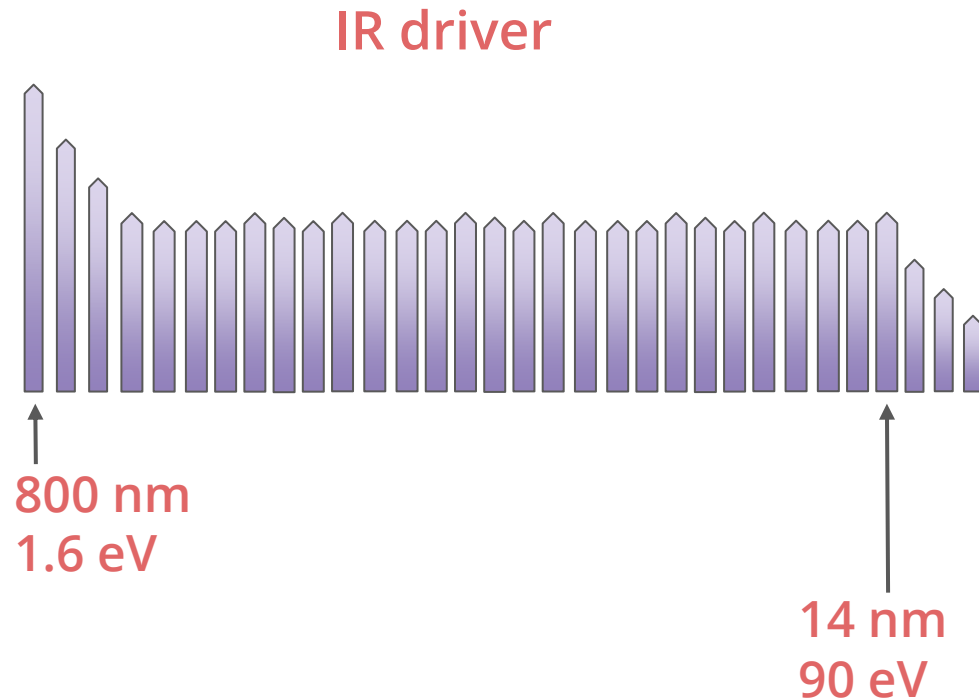
$I_p \rightarrow$ ionization energy

$U_p \rightarrow$ ponderomotive energy

$I \rightarrow$ peak intensity

$\lambda \rightarrow$ driving wavelength

J. Krause, et al. *Phys. Rev. Lett.* (1993)



High harmonic emission towards the X-ray regime

The cutoff energy law: $E_c \approx I_p + 3.17 U_p$, $U_p \propto I \lambda^2$

Scaling with the driving wavelength (or energy):

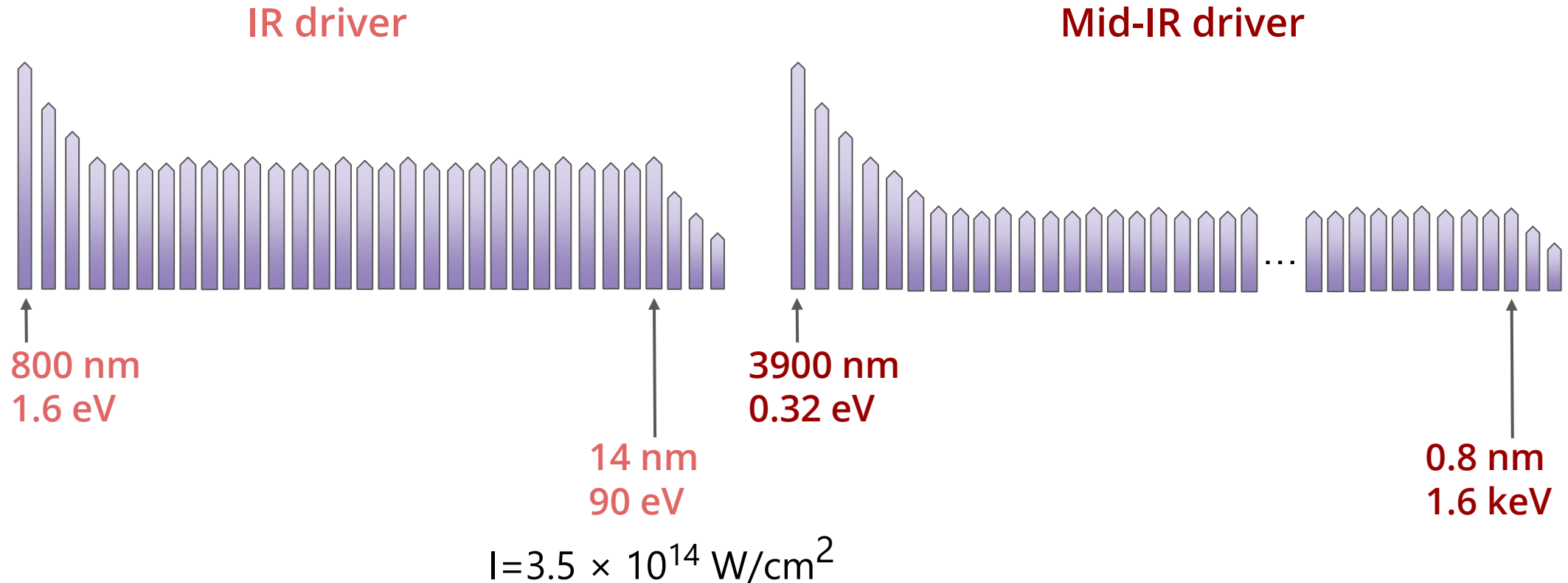
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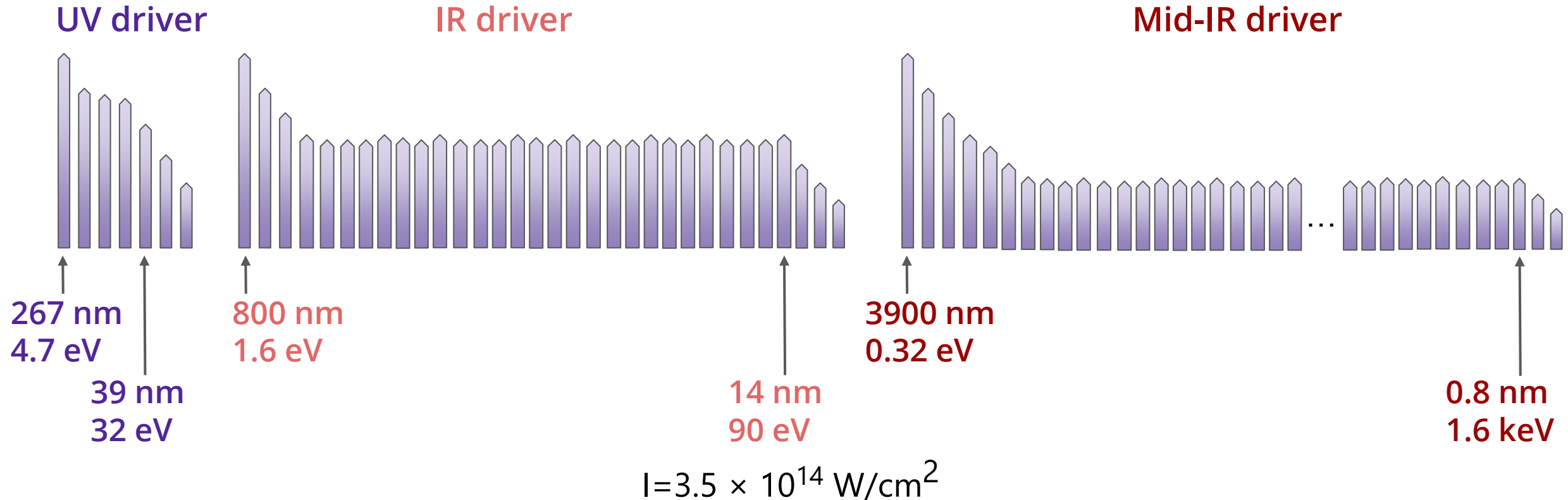
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J. Krause, et al. *Phys. Rev. Lett.* (1993)



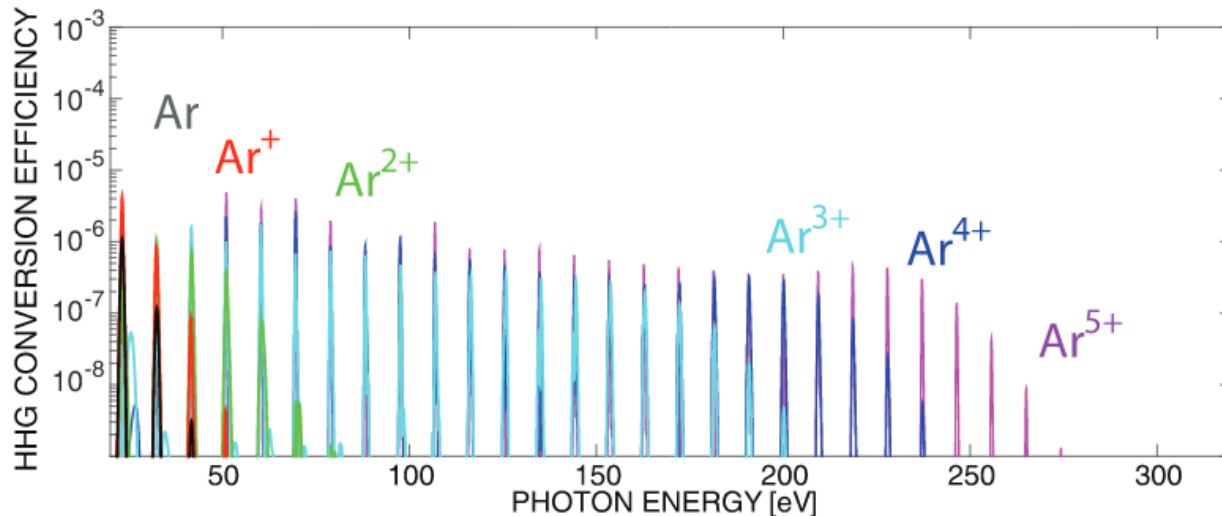
High harmonic emission towards the X-ray regime

The cutoff energy law: $E_c \approx I_p + 3.17 U_p$, $U_p \propto I \lambda^2$

Scaling with the driving wavelength (or energy):

UV driver (267 nm, 4.65 eV)

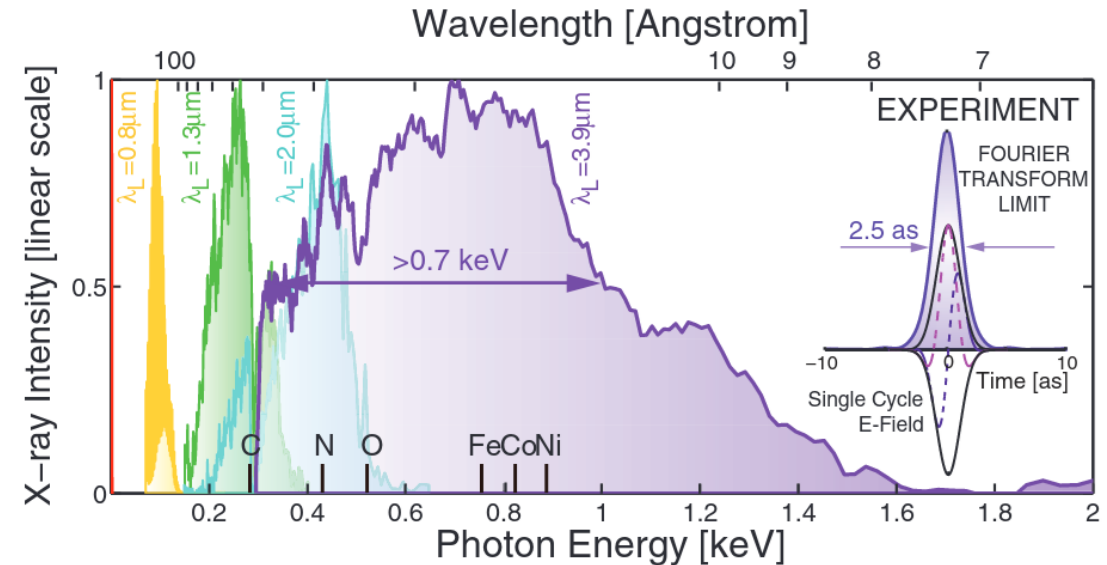
High brightness



D. Popmintchev *et al.* *Science* (2015)

Mid-IR driver (3900 nm, 0.32 eV)

High photon energies



T. Popmintchev *et al.* *Science* (2012)

Harmonics from EUV pulses?

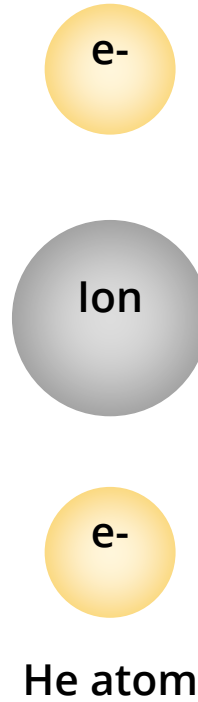
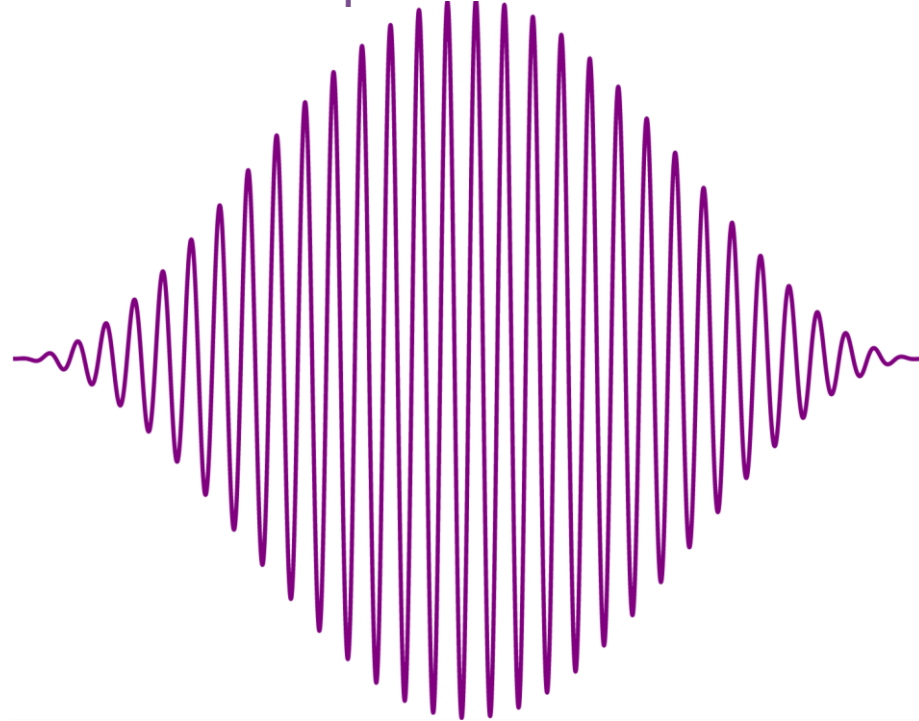


Extreme regime

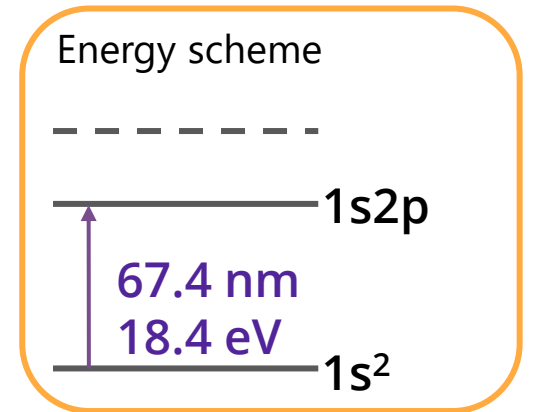
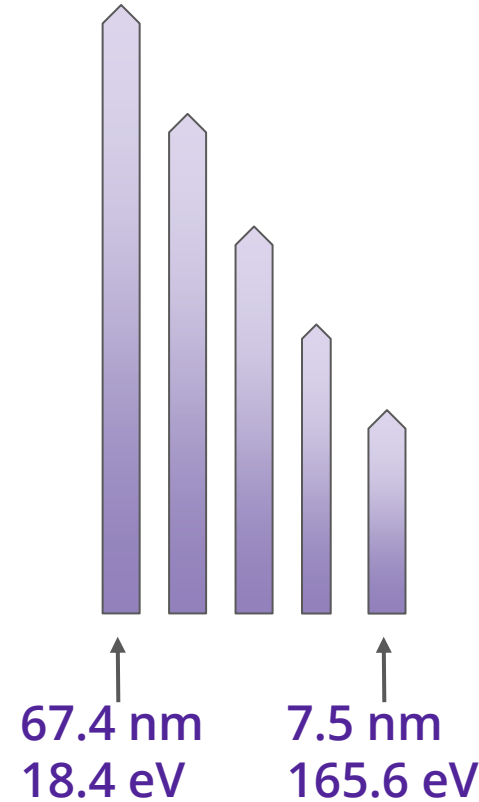
- High peak intensities (8.8×10^{13} , 8.8×10^{15}) W/cm²
Barrier supp. of He atom at 1.4×10^{15} W/cm²
- Driving energies in the range of atomic transitions
- Driving energies $\sim 0.75 I_p$
- Keldysh parameter in the range (1.8, 18.1) .

HHG **resonantly** driven by an EUV pulse

EUV femtosecond pulse

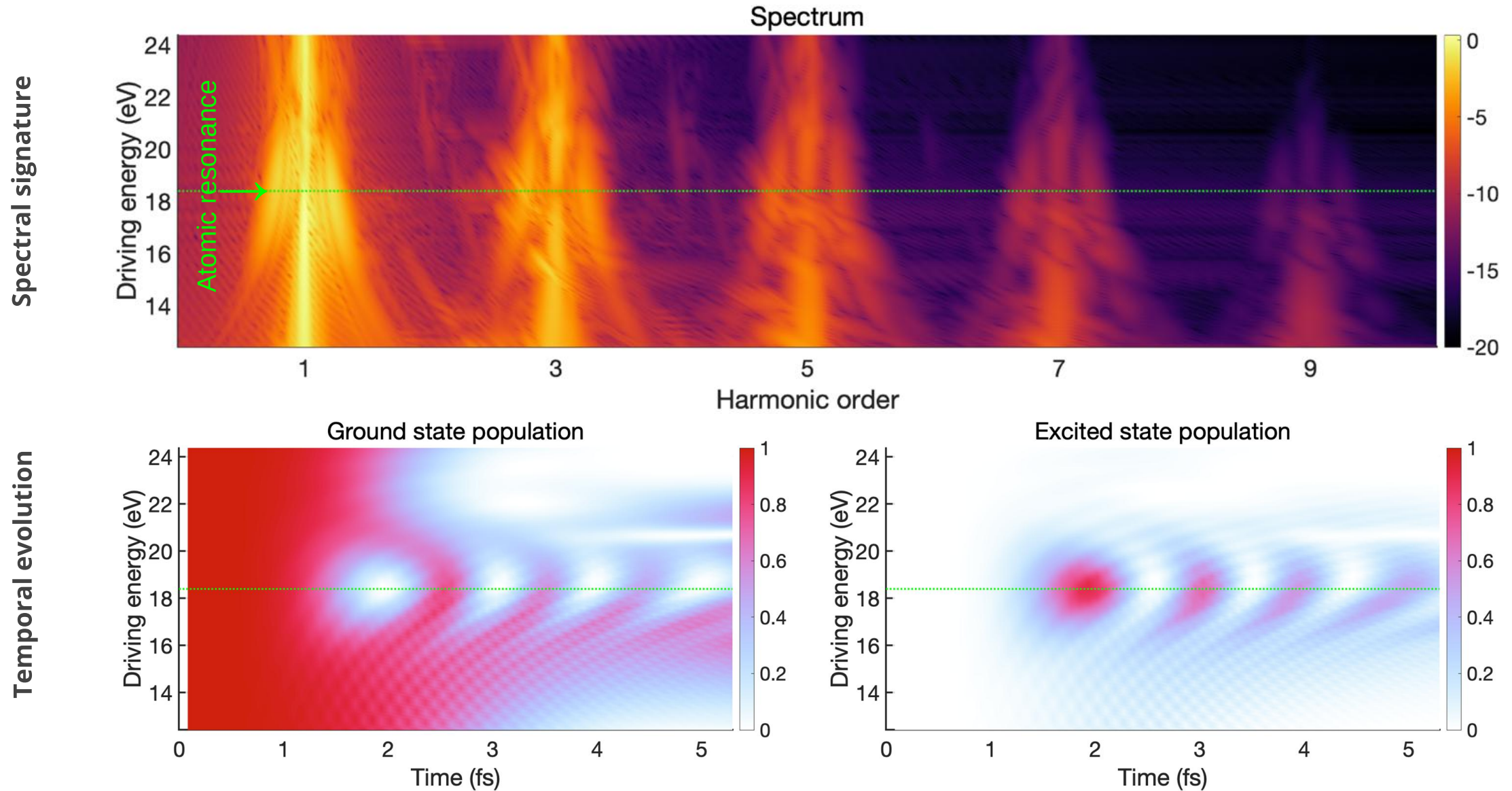


He atom



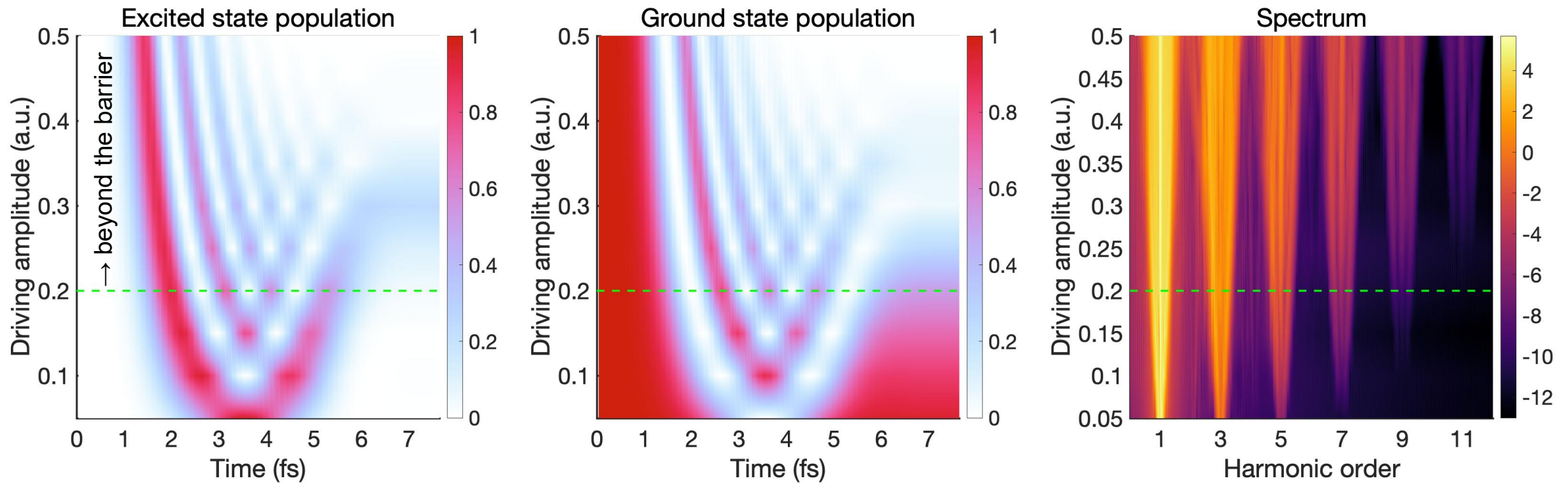
This scheme can be implemented by
generating harmonics from harmonics!

High harmonic generation resonantly driven by EUV pulses

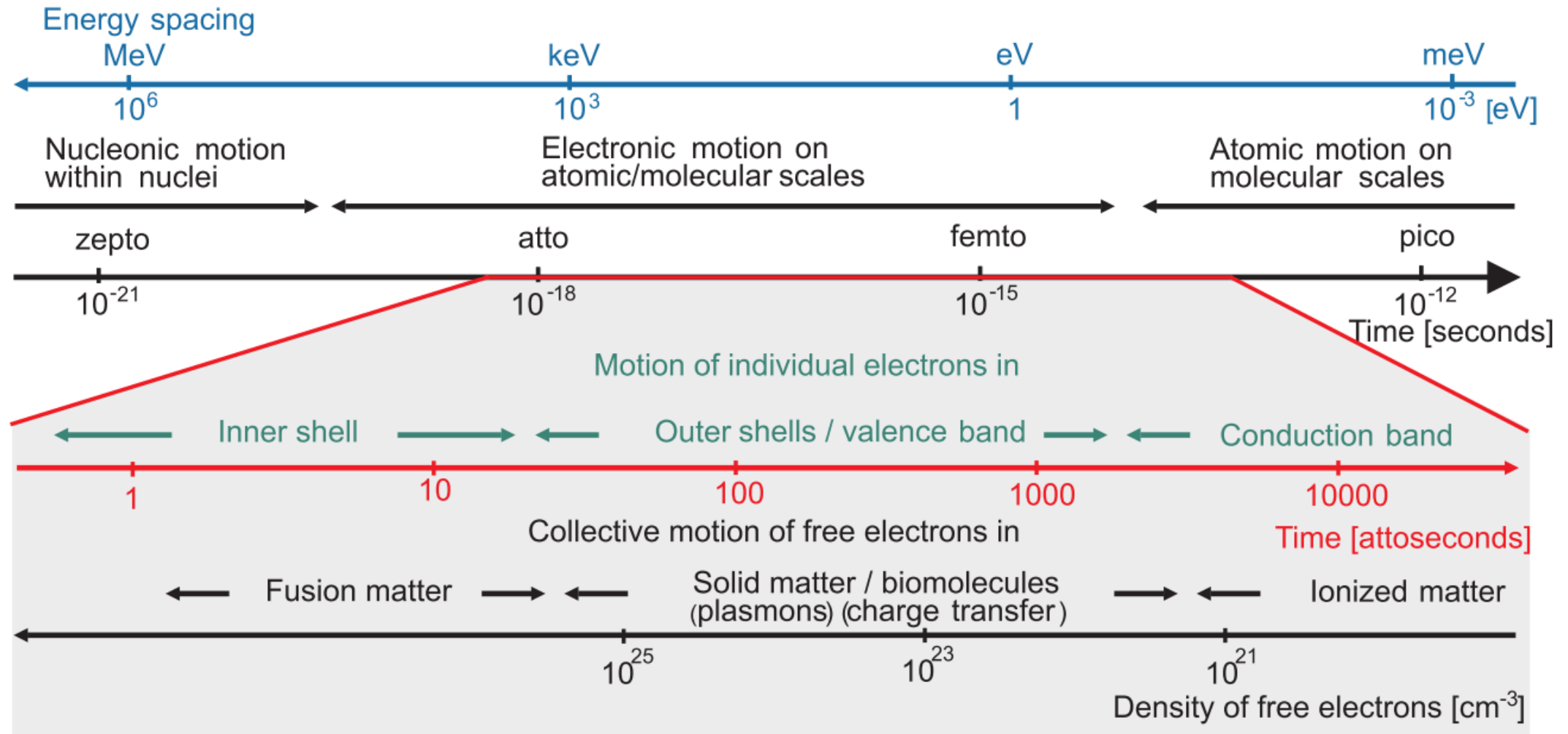


Attosecond resonant dynamics at high intensities

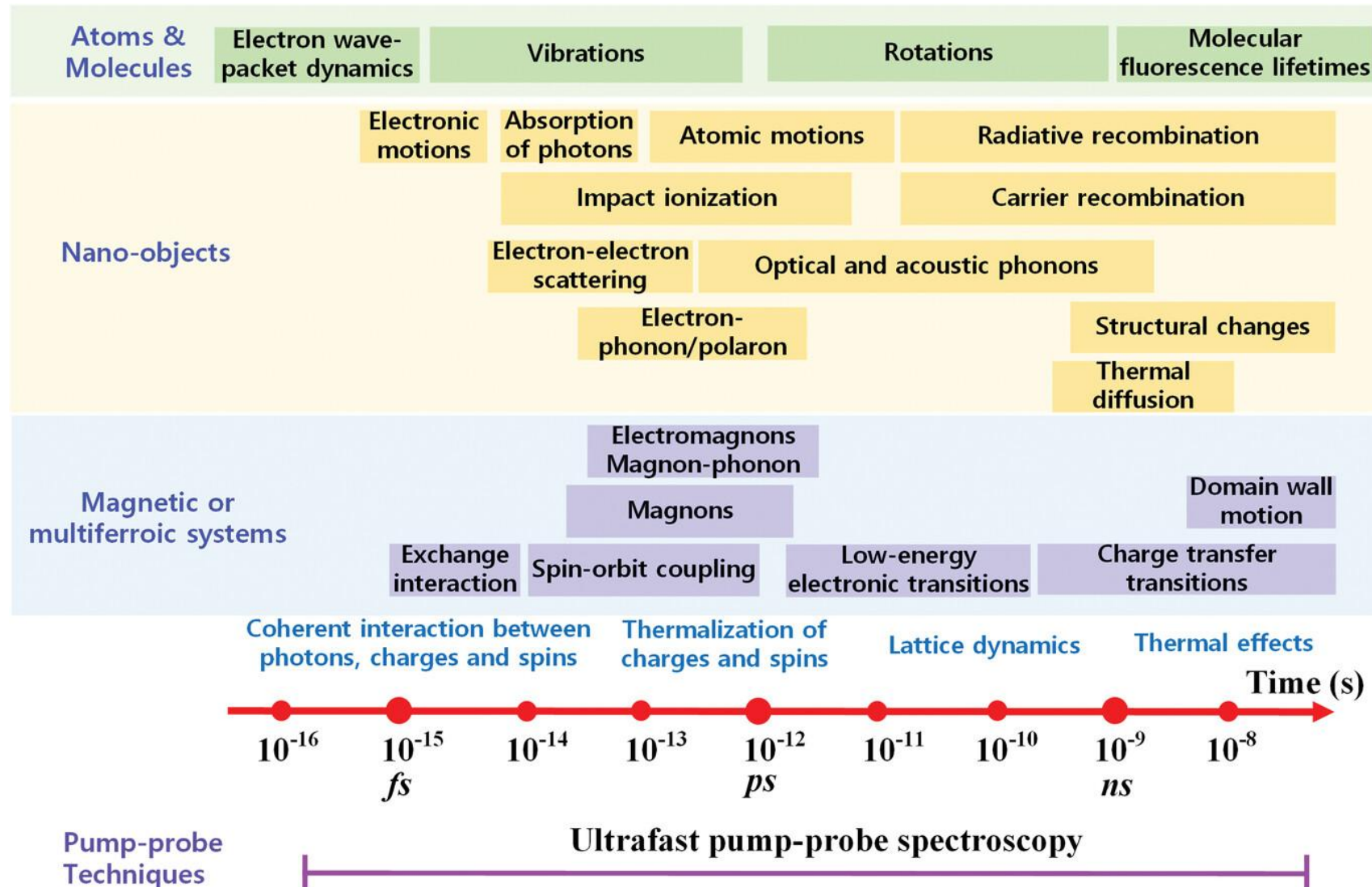
Higher field amplitudes drive faster Rabi oscillations occurring at femtosecond-to-attosecond time scales



Timescales of ultrafast dynamics in matter



Timescales of ultrafast dynamics in matter



Exercise: Provide a classical estimate of the revolution period of an electron in the Bohr orbits of a hydrogen atom. What is the time required to orbit the ground state?

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Step 1: Fundamental Period Definition

$$T = \frac{2\pi r}{v}$$

Step 2: Angular Momentum Quantization (Bohr Condition)

$$L = mvr = n\hbar \quad v = \frac{n\hbar}{mr}$$

Step 3: Substituting v into Period Formula

$$T = \frac{2\pi r}{n\hbar/mr} \quad T = \frac{2\pi mr^2}{n\hbar}$$

Step 4: Generalizing Radius (r_n)

$$r_n = n^2 a_0$$

Step 5: Final Generalized Equation

$$T_n = \frac{2\pi m(n^2 a_0)^2}{n\hbar}, T_n = \frac{2\pi mn^4 a_0^2}{n\hbar}$$

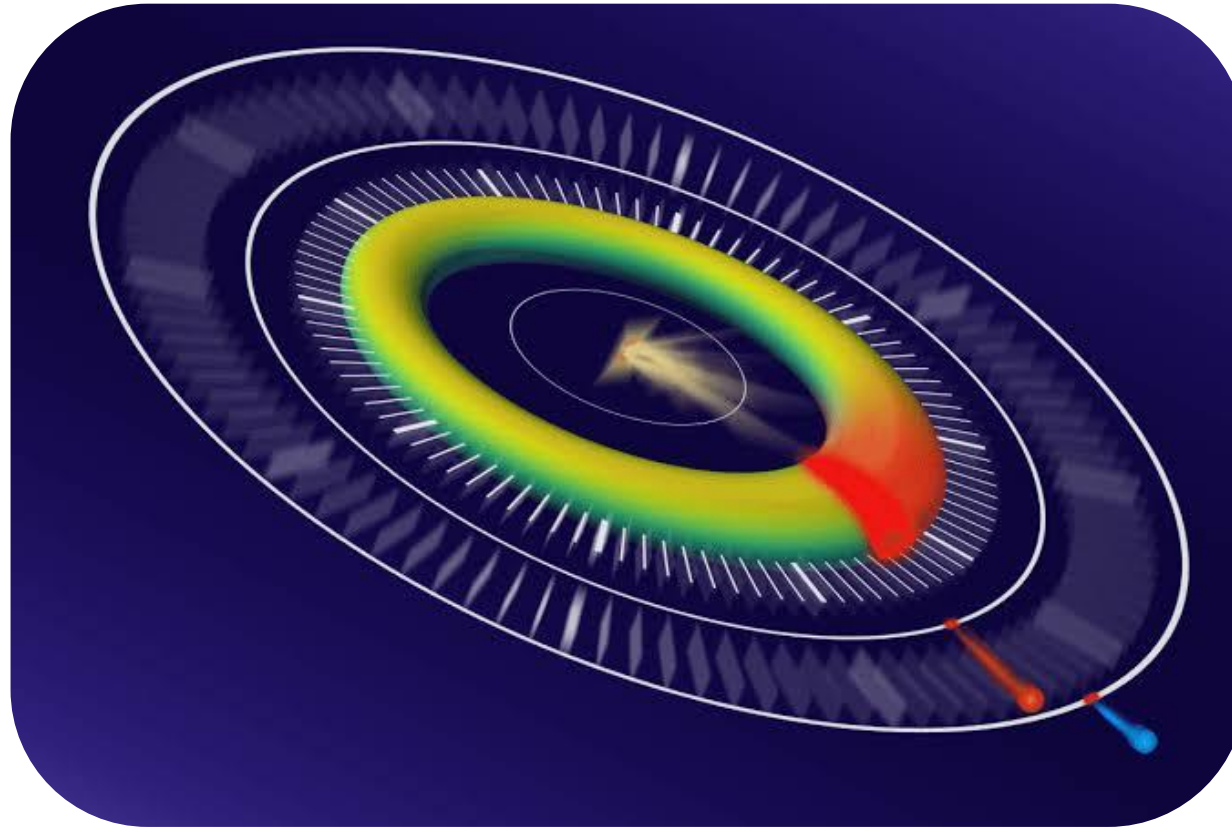
Step 6: Final Simplified Equation (Cubical Scaling)

$$T_n = \frac{2\pi ma_0^2}{\hbar} \cdot n^3$$

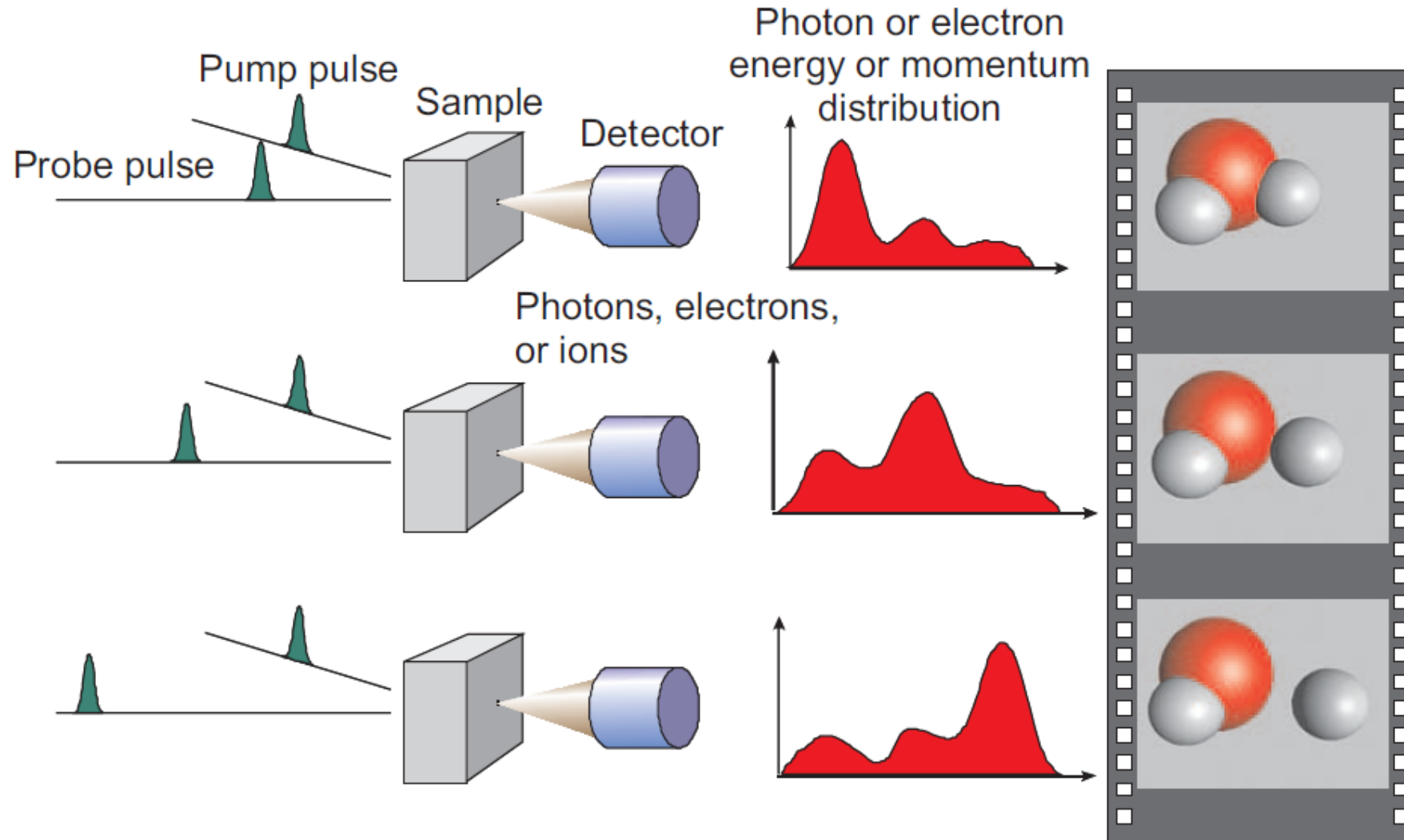
$$T_1 = 152 \text{ as}$$

Note that in reality, electrons do not follow a circular trajectory, instead they are described by electronic wavefunctions!

Attosecond-resolved spectroscopy with pump-probe schemes

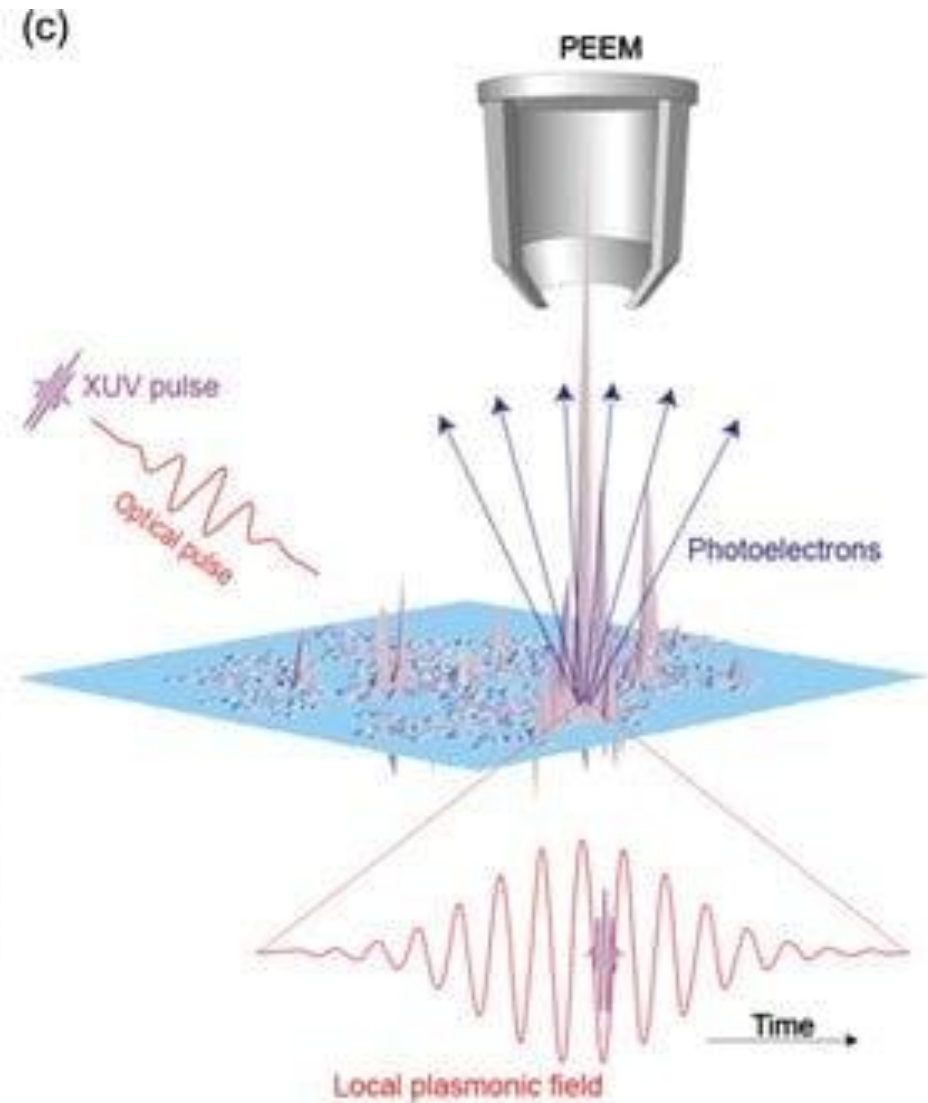
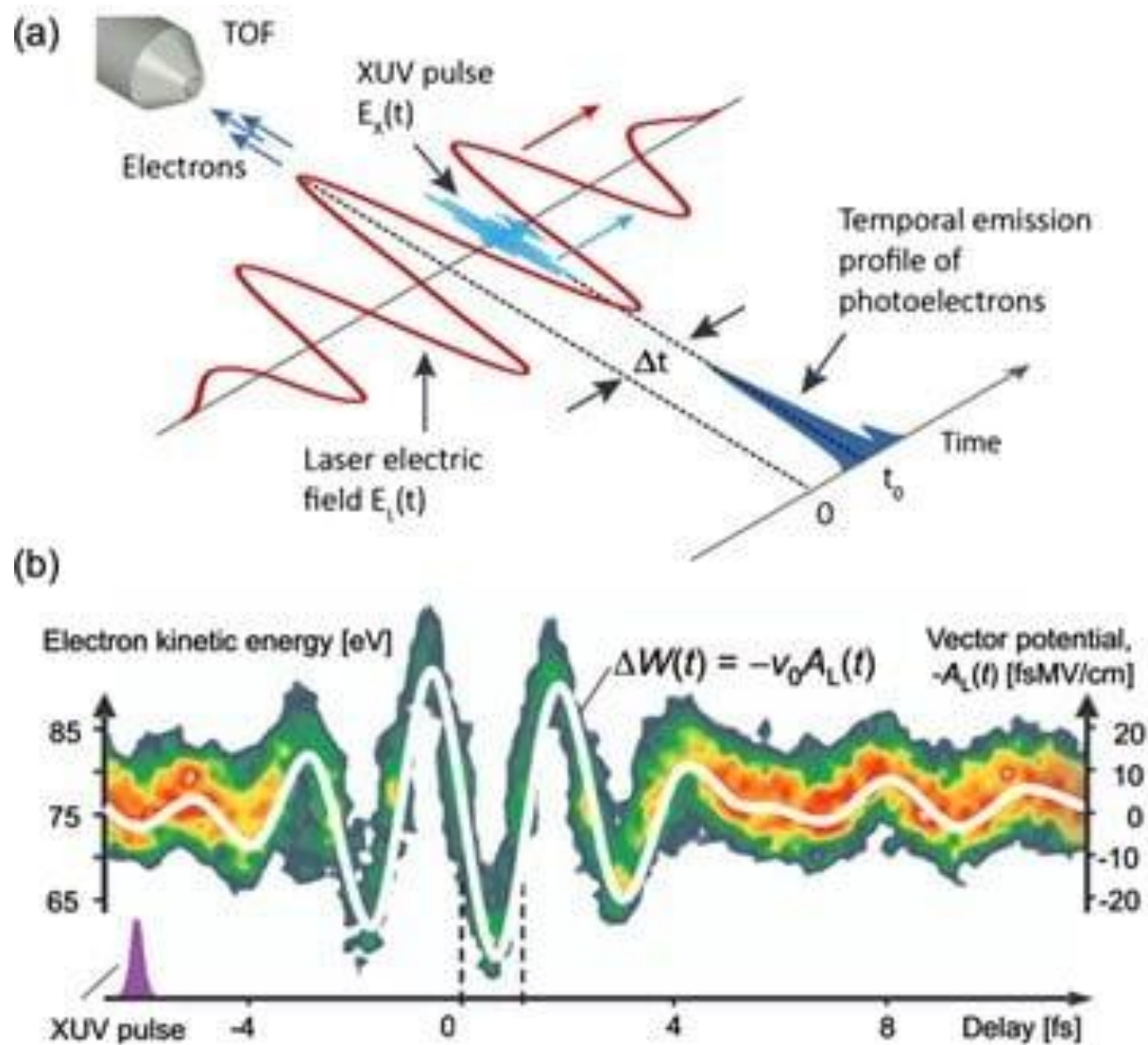


Pump-probe spectroscopy for tracking ultrafast dynamics



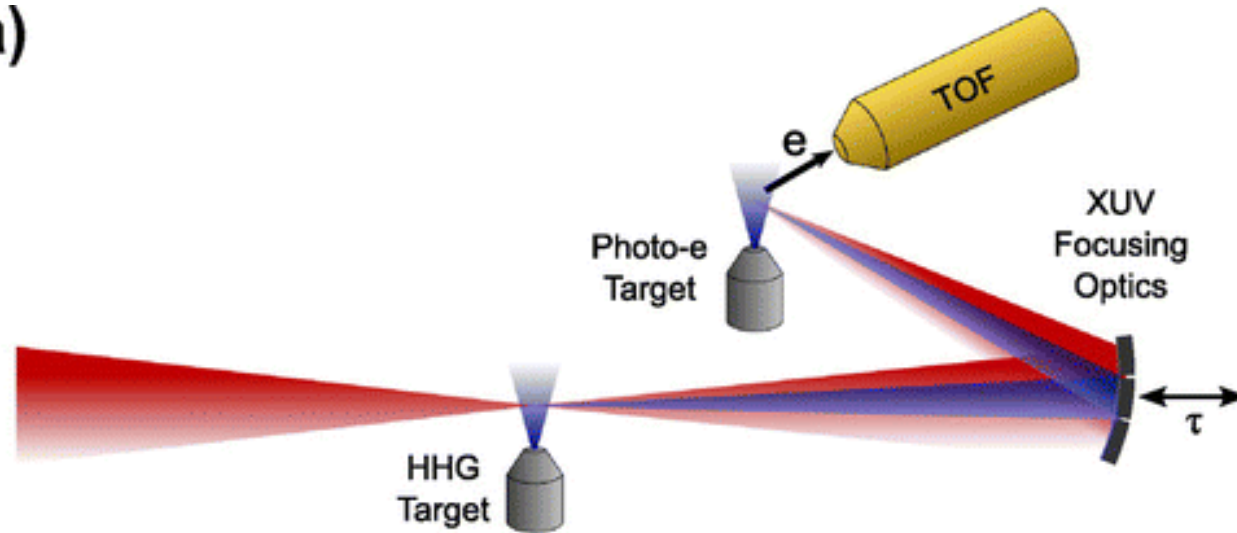
F.Krausz & M. Ivanov. *Rev. Mod. Phys* (2009)

Attosecond streaking

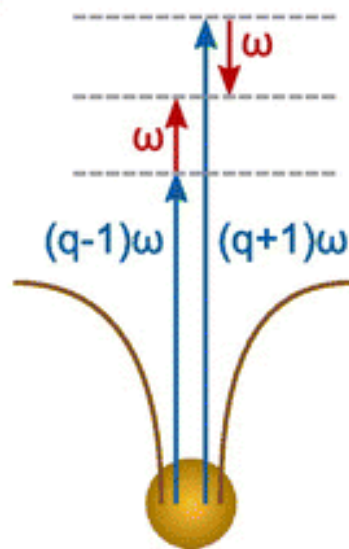


RABBIT (Reconstruction of attosecond beating by interference of two-photon transitions)

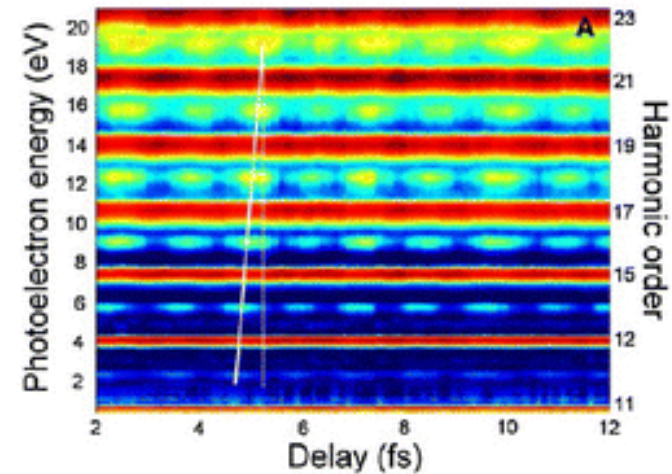
(a)



(b)

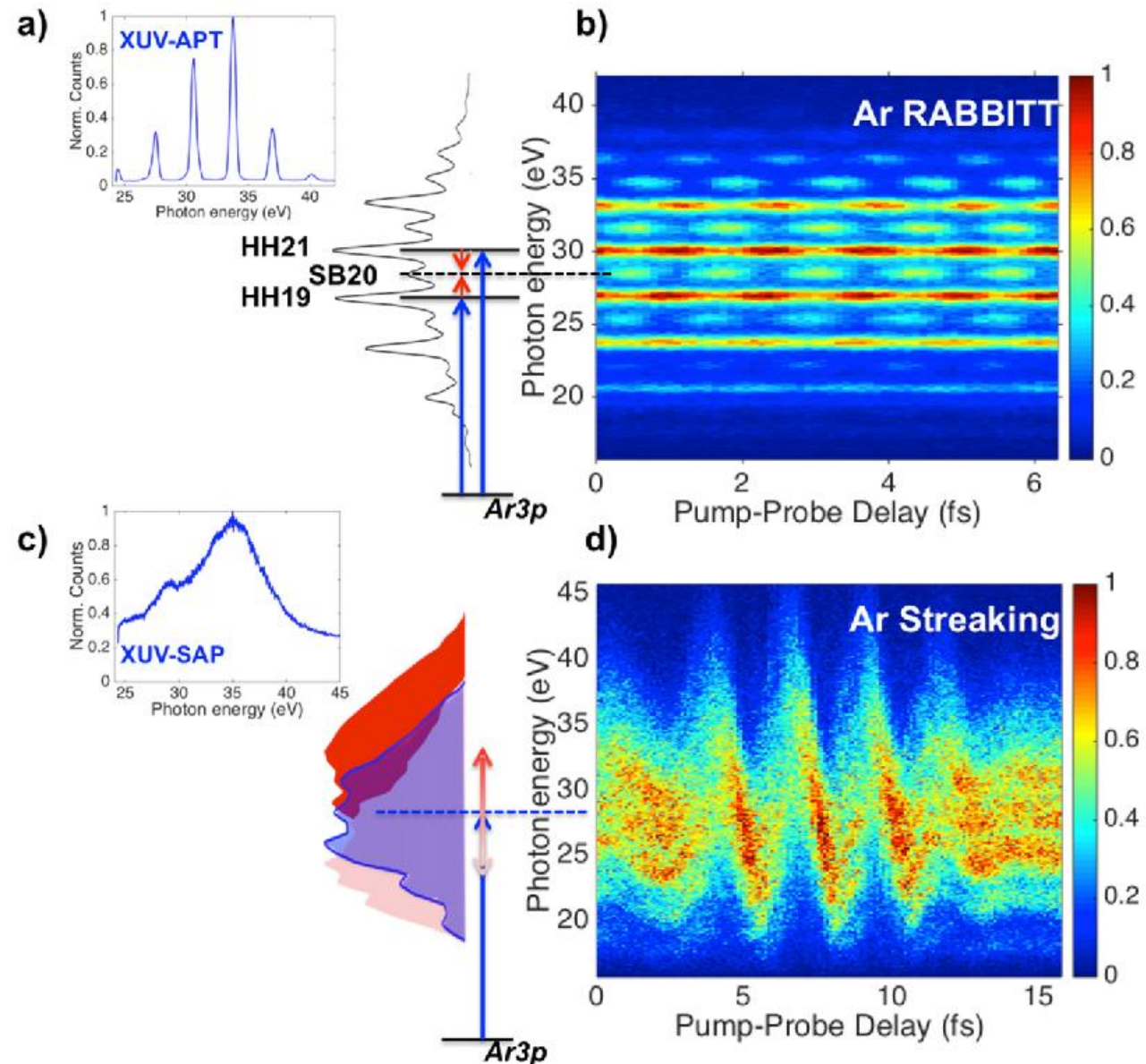


(c)



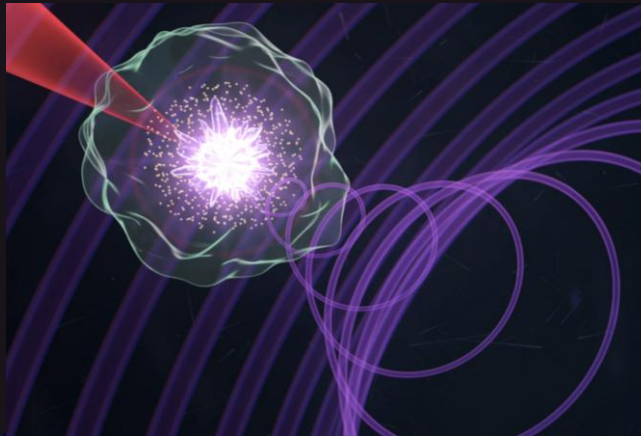
RABBIT vs Attosecond Streaking

Both techniques provide **relative timing information** between electrons originating from different initial states within the same atom or different atoms.

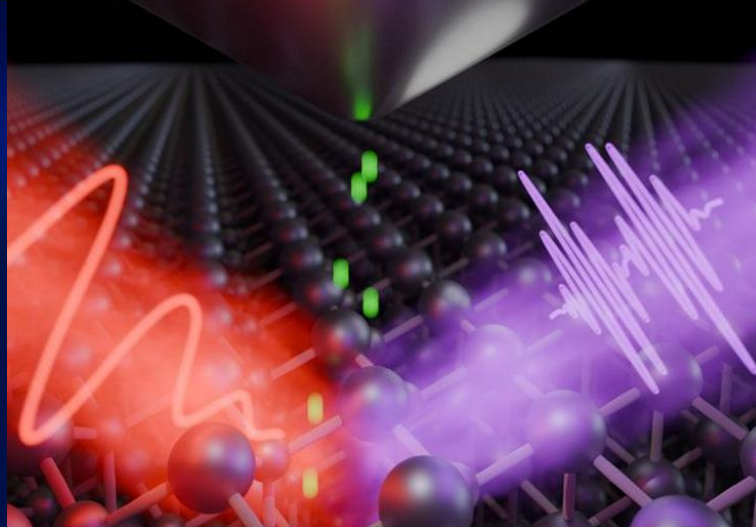
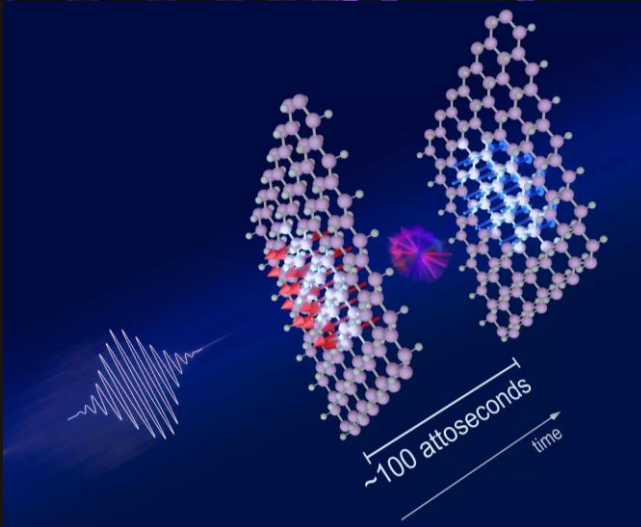
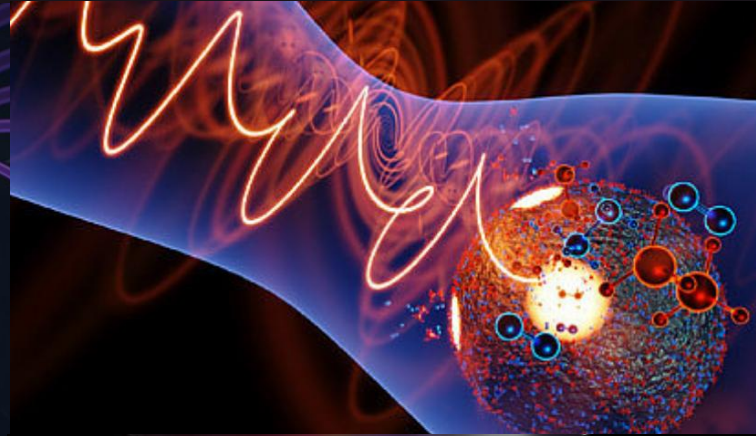


Attosecond science

Attosecond control of light



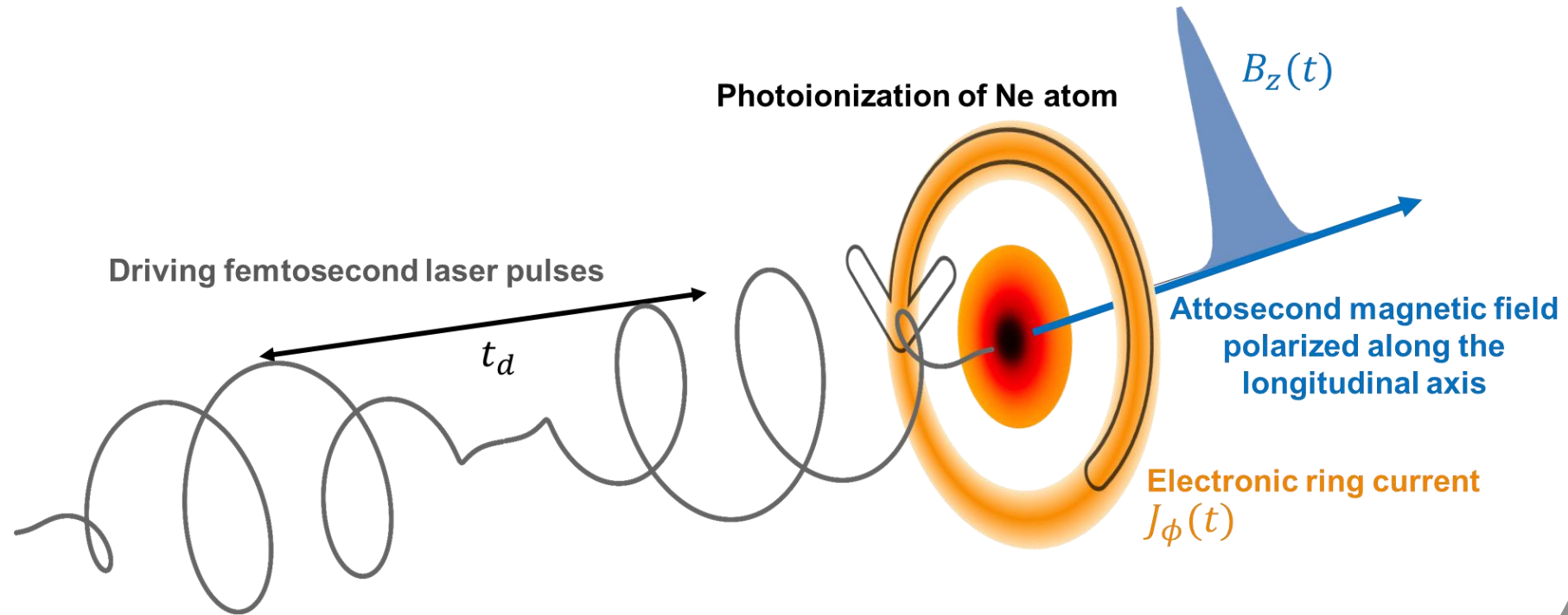
Attosecond imaging and spectroscopy



Attosecond magnetism

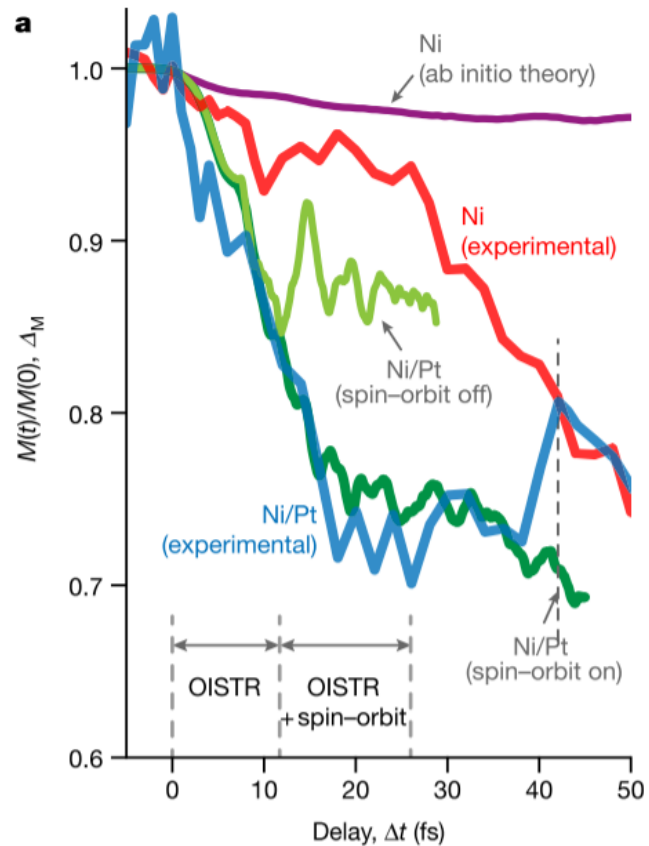
Attosecond control of matter

Generation of attosecond magnetic pulses from ring-current gating



Ultrafast magnetization dynamics

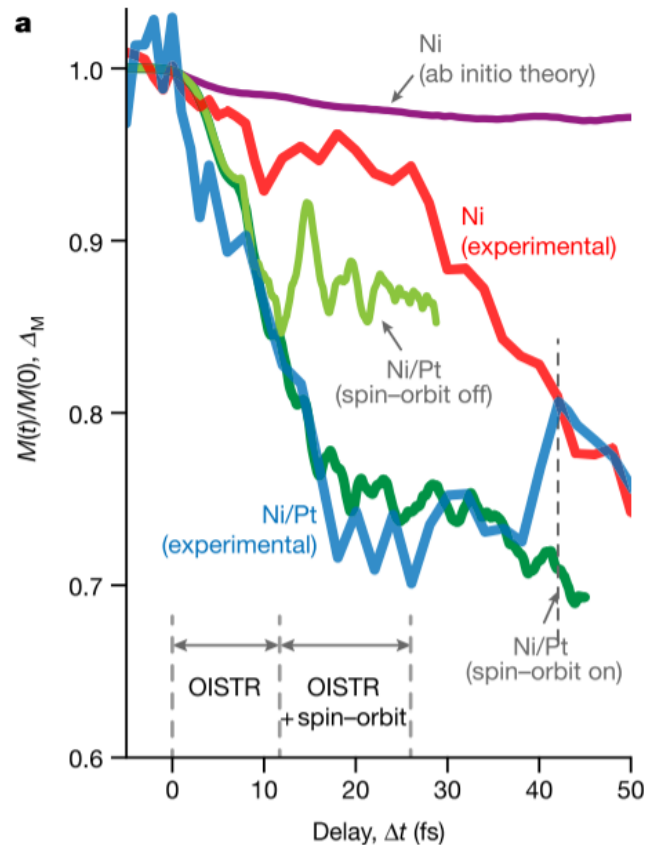
Attosecond magnetic circular-dichroism



F. Siegrist et al. *Nature* 571, 240 (2019)

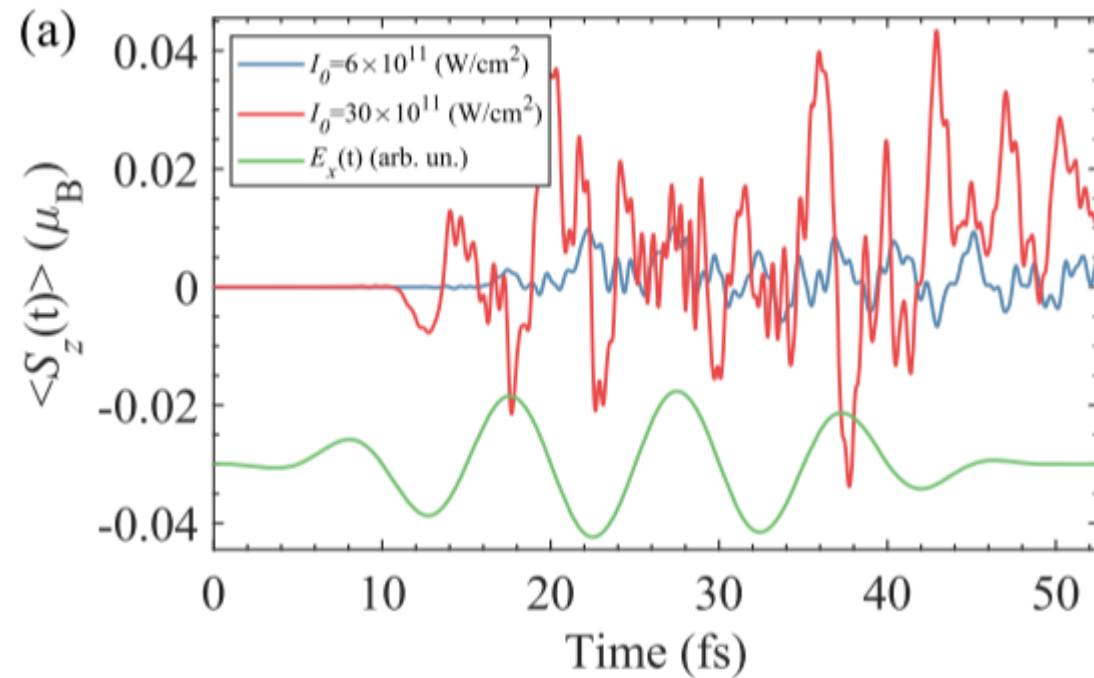
Ultrafast magnetization dynamics

Attosecond magnetic circular-dichroism



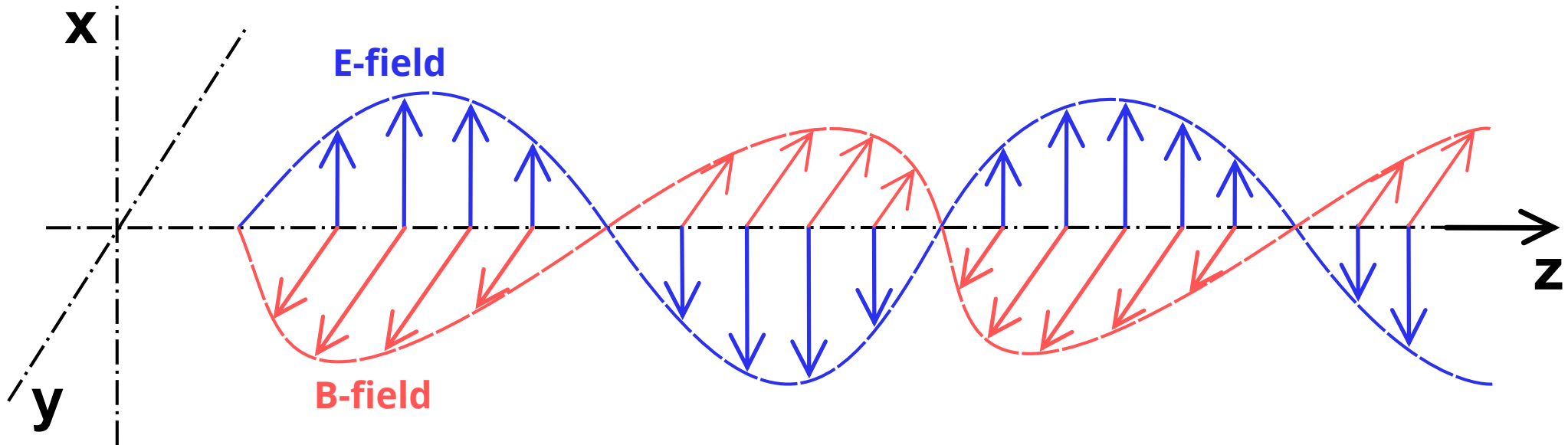
F. Siegrist et al. *Nature* 571, 240 (2019)

Theoretical prediction of faster magnetization dynamics



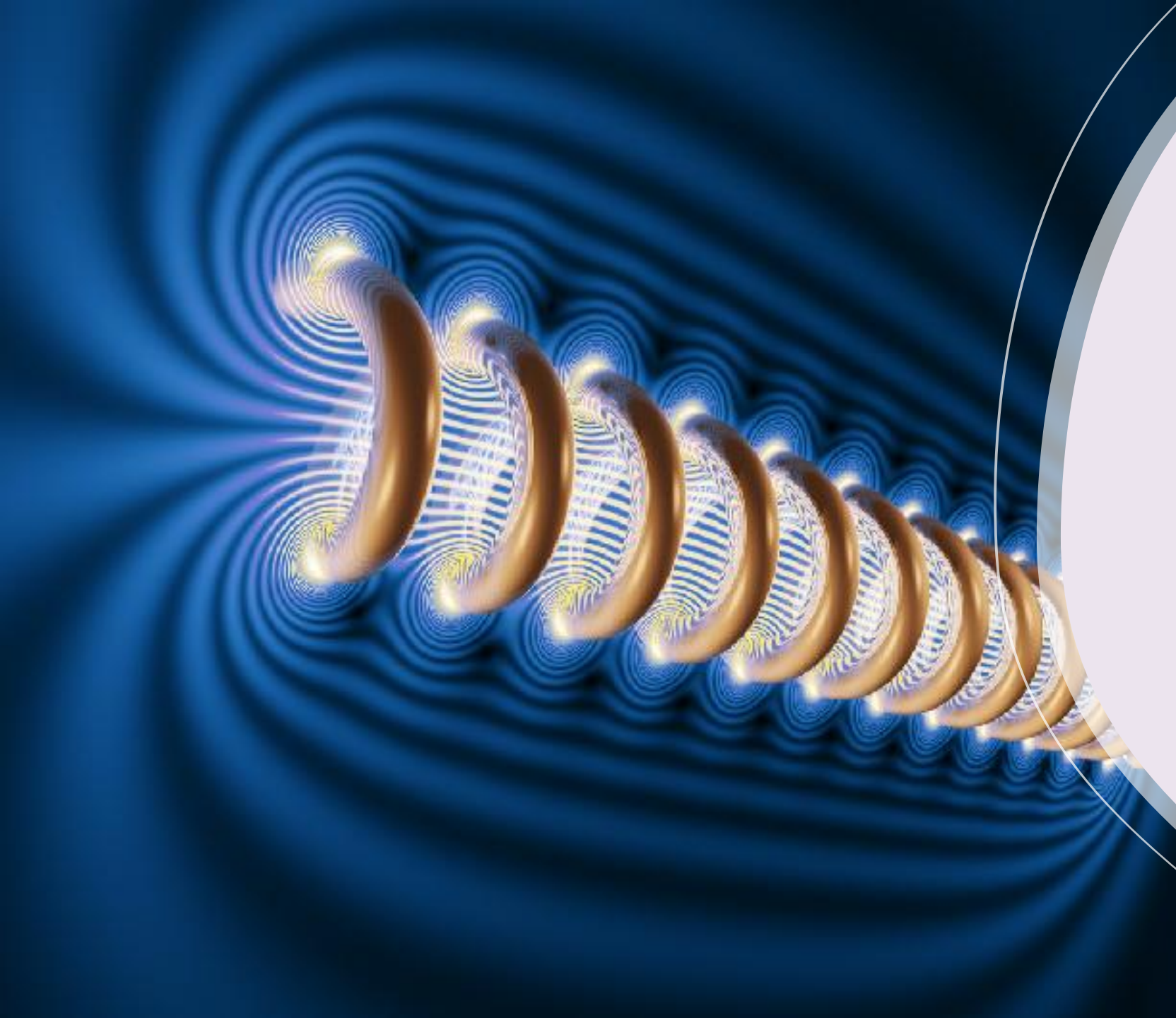
O. Neufeld et al. *Npj Comput. Mater.* 9, 39 (2023)

Electromagnetic fields



The electric field dominates the interaction of matter with electromagnetic pulses.

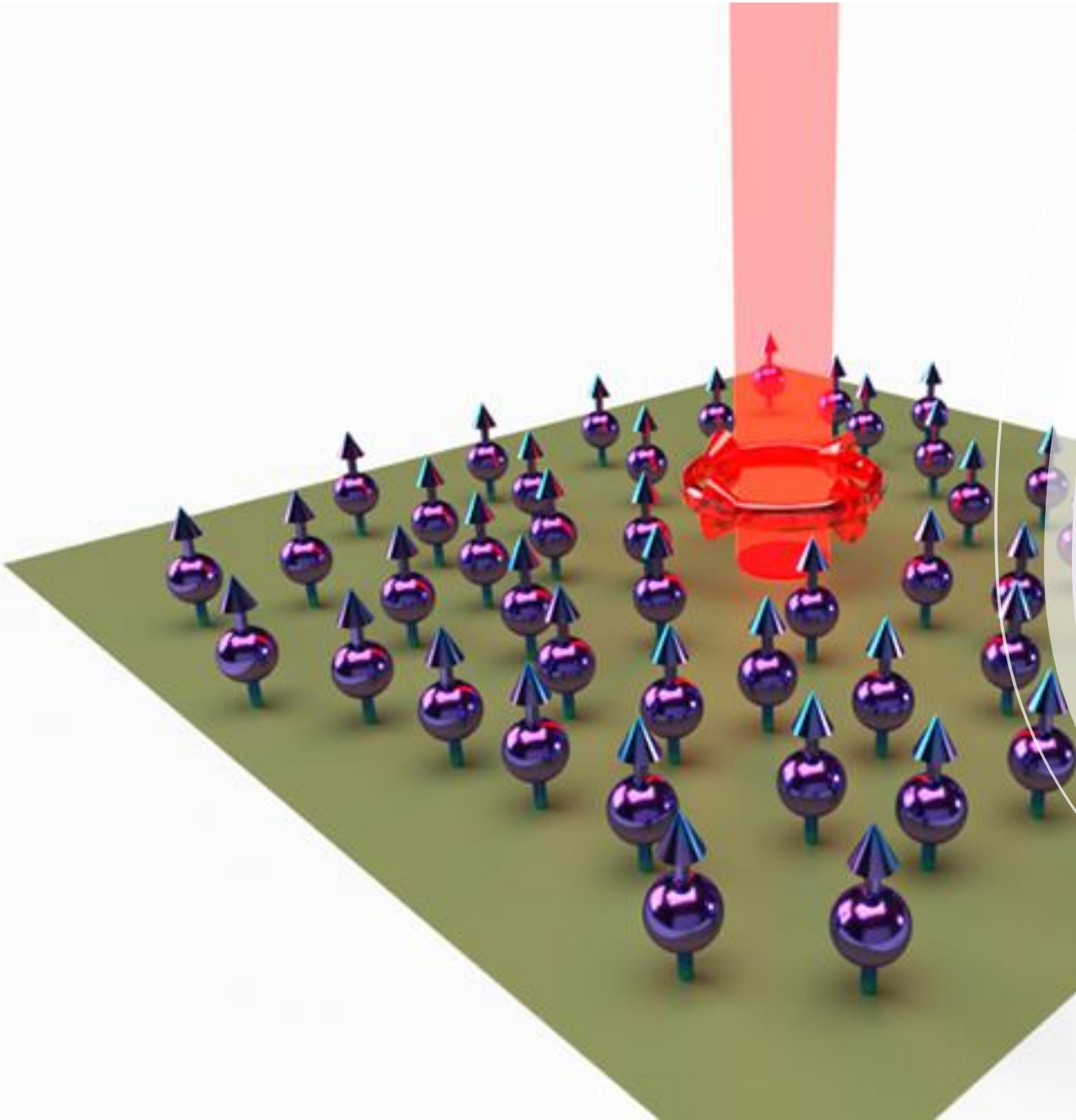
Isolated attosecond magnetic fields are desirable to explore ultrafast magnetic phenomena.



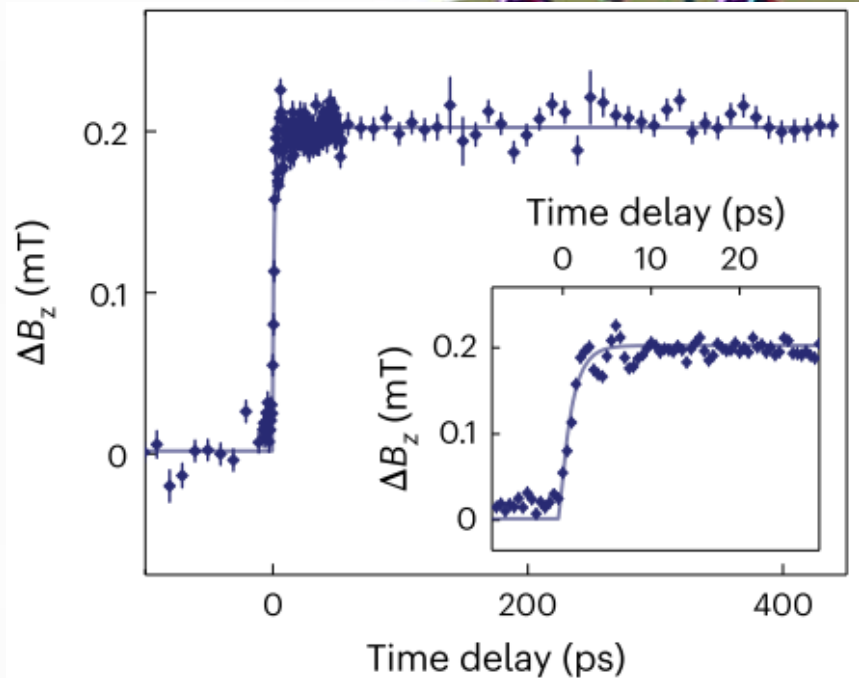
Conventional generation of magnetic fields

- Coils and solenoids can produce strong magnetic fields but not with short duration.
- Suitable for quasi-static magnetic phenomena or long-duration magnetization dynamics.

Magnetic fields from laser-induced electron currents



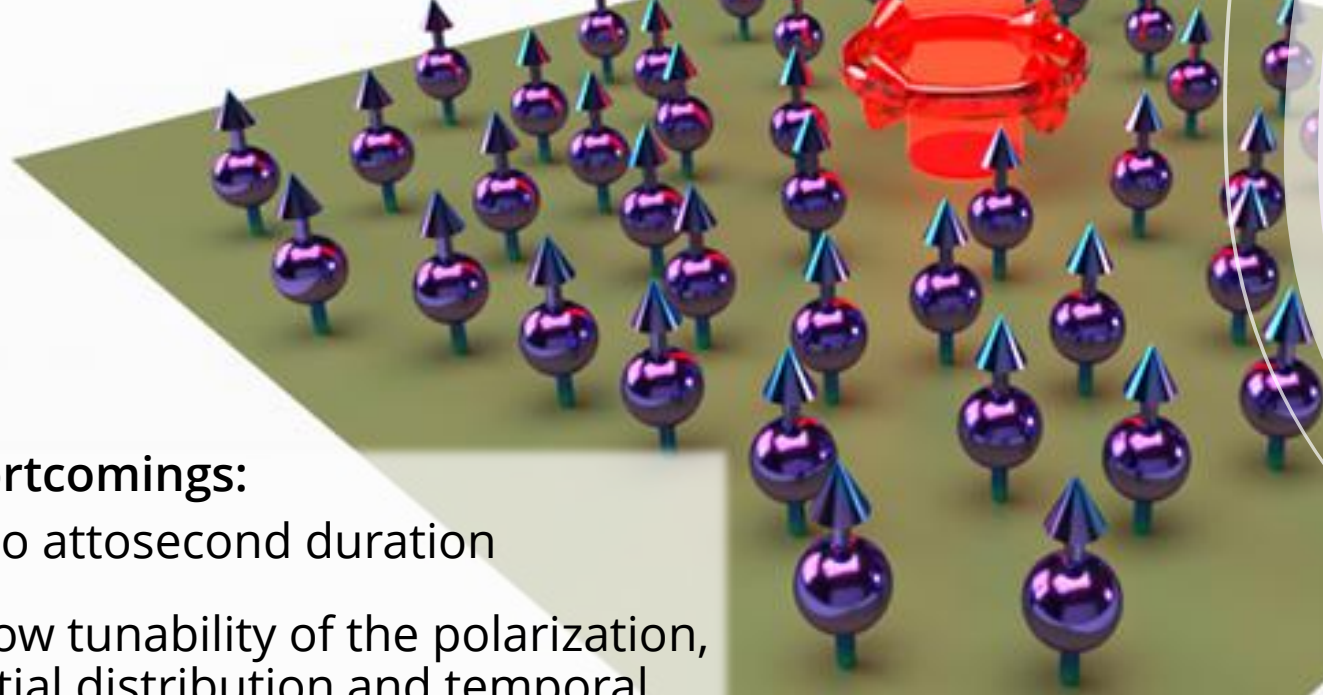
I. Barth, J. Manz. *Phys. Rev. A* 75, 012510 (2007)
V. Fedotov, *et al. New J. Phys.* 15, 113035 (2013)
K.J. Yuan, A.D. Bandrauk. *Phys. Rev. A* 92, 063401 (2015)
J. Wätzel, *et al. J. Mod. Opt.* 64, 1088 (2017)
Y. Shi, *et al. Phys. Rev. Lett.* 121,145002 (2018)
J. Guo, *et al. Phys. Rev. A* 99, 053416 (2019)
M. Blanco, *et al. ACS Photonics* 6, 38 (2019)
S. Sederberg, *et al. Nat. Photonics* 14, 680 (2020)
S. Sederberg, *et al. Phys. Rev. X.* 10, 011063 (2020)
K. Jana, *et al. Nat. Photonics* 15,622 (2021)
X. Yang, *et al. ACSNano* 16, 386 (2022)
G. De Vecchi, *et al. ArXiv* 2408.05153



Magnetic fields from laser-induced electron currents

- Different implementations in plasmas, semiconductors, nanostructures, superconductors, molecules or atoms.
- Characterized by an ultrafast turn-on, followed by a relatively long-lived magnetic field.

I. Barth, J. Manz. *Phys. Rev. A* 75, 012510 (2007)
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K.J. Yuan, A.D. Bandrauk. *Phys. Rev. A* 92, 063401 (2015)
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S. Sederberg, *et al. Phys. Rev. X.* 10, 011063 (2020)
K. Jana, *et al. Nat. Photonics* 15, 622 (2021)
X. Yang, *et al. ACS Nano* 16, 386 (2022)
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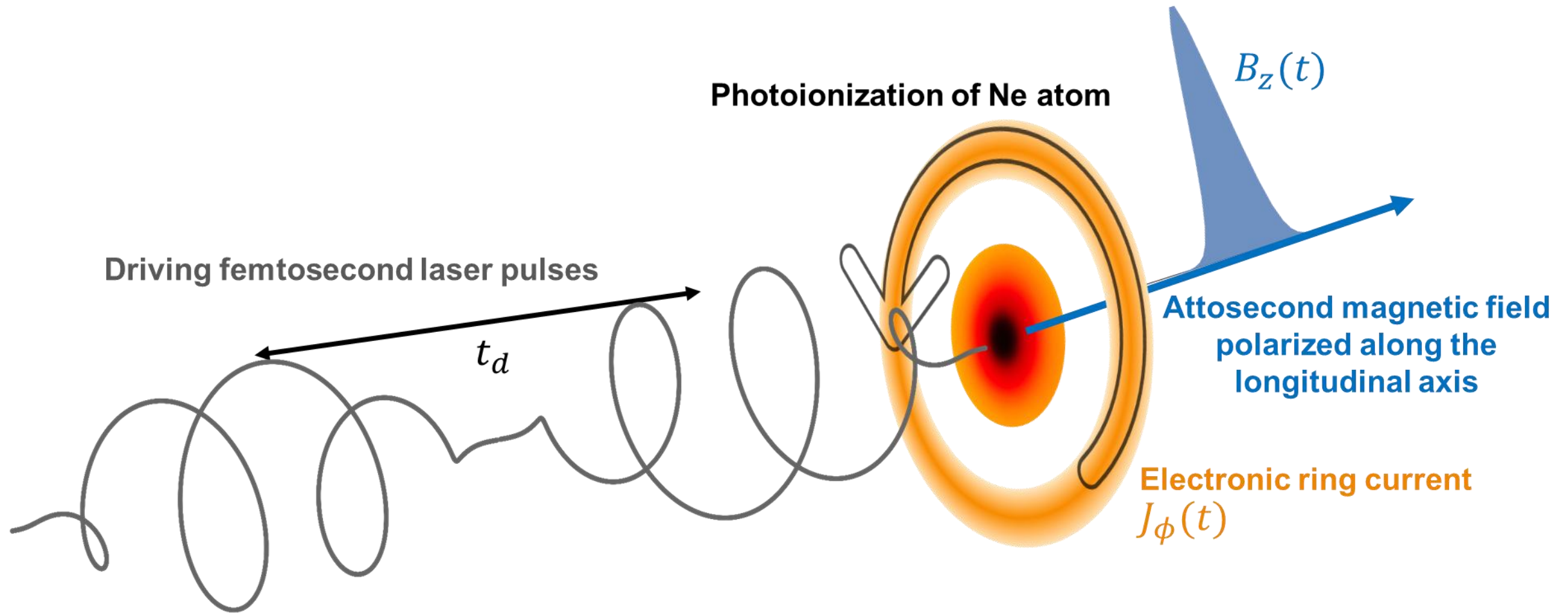
Magnetic fields from laser-induced electron currents

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Shortcomings:

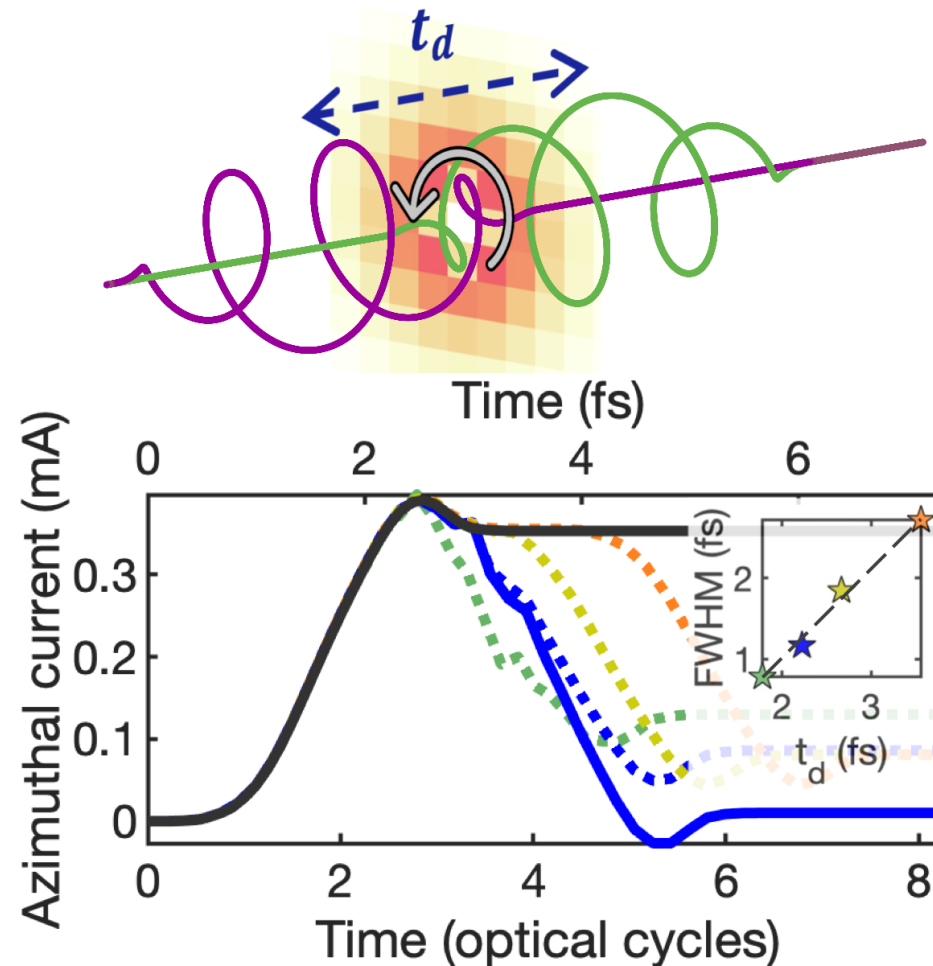
- No attosecond duration
- Low tunability of the polarization, spatial distribution and temporal waveform

Our idea of ring-current gating



Tunable attosecond magnetic field waveforms via ring-current gating

Two gating femtosecond laser pulses



A. de las Heras *et al.* *J. Phys. Chem. Lett.* (2023)

Autoevaluation form of SESSION 1

<https://forms.gle/oEZfoHGN5pBhS93XA>

